



The impact of network connectivity on factor exposures, asset pricing, and portfolio diversification[☆]

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ABSTRACT

This paper extends the classic factor-based asset pricing model by including network linkages, leading to a network-augmented linear factor model. This extension of the model allows a better understanding of the determinants of systematic risk and shows that cross-sectional risk premia can be estimated more precisely. Moreover, we demonstrate that in the presence of network links a misspecified traditional linear factor model presents residuals that are correlated and heteroskedastic. We support our claims with an extensive simulation experiment and real data.

1. Introduction

The term “systematic risk” is a well-established concept that derives from the seminal work on portfolio choice proposed by Markowitz (1952) and extended in a general equilibrium framework by Sharpe (1964), Lintner (1965, 1975) and Mossin (1966), and in the Arbitrage Price Theory model by Ross (1976). It refers to the risk to which an investor in a well-diversified portfolio is exposed, which stems from the dependence of the returns on common factors. However, as stressed by Cochrane (2011), there is a need for a better understanding of the determinants of systematic risk.

In this paper, we provide a unique framework for systematic risk and network connectivity, and we estimate the feedback between network exposures and common factors and their impact on the factor risk exposures and risk premia of stock returns.

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A growing literature investigates the role of interconnections between different firms and sectors as a potential mechanism for the propagation of shocks throughout the economy. Acemoglu, Carvalho, Ozdaglar, and Tahbaz-Salehi (2012) use network structure to show the possibility that aggregate fluctuations may originate from microeconomic shocks to firms. Herskovic, Kelly, Lustig, and Van Nieuwerburgh (2020) show how stock firm volatility is related to customer–supplier connectedness. Billio, Gray, Getmansky, Lo, Merton, and Pelizzon (2014) use contingent claim analysis and network measures to highlight interconnections between sovereigns, banks, and insurance. There are several other contributions to the literature on network analysis: see Billio, Getmansky, Lo, and Pelizzon (2012), Diebold and Yilmaz (2015), Hautsch, Kyj, and Oomen (2012), Hautsch, Kyj, and Malec (2015), Barigozzi and Brownlees (2019), Ozdagli and Weber (2017), Fernandez (2011), Kou, Peng, and Zhong (2017), and Buraschi and Tebaldi (2017). Network interconnections and the effects called network externalities- that arise from small and local shocks that can become big and global are a possibility discarded in standard asset pricing and macroeconomic models due to a “diversification argument”. As argued by Lucas (1977), among others, microeconomic shocks will average out and thus have only negligible aggregate effects. Similarly, they will have little impact on asset prices. However, there is also a growing literature on the role of sectorial shocks in macrofluctuations. Examples include Horvath (1998), Dupor (1999), Shea (2002), and Acemoglu et al. (2012). Moreover, Ang, Hodrick, Xing, and Zhang (2006), among others, show that idiosyncratic volatility risk is priced in the cross-section of expected stock returns, a regularity that is not subsumed by size, book-to-market, momentum, or liquidity effects. From a theoretical point of view, Wagner (2010), Ozsoylev and Walden (2011), Buraschi and Porchia (2012), and Branger, Konermann, Meinerding, and Schlag (2021) arrive at similar conclusions. Ahern (2013) empirically documents a positive market price of centrality-i.e., that more central assets earn higher expected returns. Buraschi and Tebaldi (2017) focus on vulnerability to network risk and show that it is associated with a risk factor impacting the cross-section of expected returns. Herskovic (2018), using an Input/Output (I/O) matrix in his general equilibrium model, shows that network concentration (output share), sparsity (strength links among nodes), and productivity (that arises endogenously), are relevant characteristics affecting the stochastic discount factor.

The contribution of this paper is to propose a modelling framework in which network connectivity and common factor risks coexist. The proposed model is a variation on the traditional Capital Asset Pricing Model (CAPM) or Arbitrage Pricing Theory (APT) framework, in which networks are used to infer the exogenous and contemporaneous links across assets. By using our network-augmented linear factor model, we can disentangle direct exposures of a single stock to common factors from the indirect exposure to the common factors that arise from network interconnections. We also provide a number of generalizations to our approach to make it more flexible and coherent with the empirical evidence, for instance allowing for asset-specific reaction to network links and introducing time variation into networks. From the economic point of view, network connectivity and common factor exposures can co-exist as the former might represent the interdependence between companies arising, for instance, from trade links, while the latter represent the exposures of companies to macro-drivers. The joint presence of these two sources of returns variation allow for a more detailed analysis on the systematic risk factor exposures and, at the same time, allow improving the inferential procedures on the associated premiums.

Building on the network-augmented linear factor model, we provide a number of theoretical elements and pieces of empirical evidence based both on a simulation framework as well as on empirical data. First, focusing on the dynamics of returns and the exposure to common factors, we show that the presence of asset interconnection acts to inflate the exposure to common risk sources. Moreover, we are able to disentangle the structural exposure to common factors, which is present even in the case of no network connections, from the exposure associated with network links. A similar argument applies to the shocks impacting an asset return, whereby network relations expose assets to other assets' shocks. From a risk perspective, our approach allows us to decompose the risk of a single asset (or a portfolio) into four components: (i) the systematic component, (ii) the idiosyncratic component, (iii) the impact of the asset interconnections on the systematic risk component-that is, the contribution of network exposure to the systematic risk component and (iv) the effect of interconnections between the idiosyncratic risks on the systematic risk component-that is, the amplification of idiosyncratic risks that generates systematic/nondiversifiable risk. Building on this result, we show how diversification benefits are reduced in the presence of network connectivity. Moreover, by combining the return dynamics with the variance decomposition, we can verify that our model is consistent with the presence of correlation and heteroskedasticity among traditional linear-factor-model residuals, thus providing a rationale for the empirical evidence found in the literature.

The simulation analysis allows us to disentangle the error estimation of linear factor models that ignore the presence of network connections. In particular, we show that, when asset returns are significantly related to network interconnections, the factor loading estimation of common factors is largely misspecified if the estimation is based on a traditional linear factor model. Moreover, the residuals' correlations start drifting away from zero if network connections are ignored in the model estimation.

We also evaluate the impact of networks on the estimation of risk premia and show that the premium estimated by our approach and by a traditional linear factor model are equivalent in the long run (under some assumptions on the evolution of the network over time). Point estimates of the risk premia coincide, while our approach provides a more precise evaluation of the standard errors (thus increasing estimators' efficiency) and lead to more appropriate evaluation of the risk premia statistical significance. In addition, our approach allows local (conditional) expected returns to change, according to changes in the network structure, and thus leads to price changes even if the risk premia are time invariant. We thus introduce a generalization of the approach proposed by Fama and MacBeth (1973) for the estimation of risk premia in the presence of time-varying network dependence among assets. The simulation evidence suggests that our estimation approach provides more concentrated risk premia with effects on the evaluation of the risk premia's statistical significance. The empirical evidence on real data supports our view, showing, in particular, that trade-related network interdependence co-exist with the exposure to common factors. Moreover, on the one side, a traditional linear factor model, for our series and sample, does not provide evidences of statistically significant risk premiums, while, on the other side, our approach leads to significant risk premiums, thus leading to more accurate predictions for the cross-section of returns.

The remainder of the paper is organized as follows. Section 2 presents our model combining network links and factor exposure, while Section 3 introduces a set of generalizations, making the model more flexible. Section 4 describes the estimation methodology for the model augmented with the network links. Section 5 presents the simulation analysis, Section 6 shows the results of the empirical analysis on the 48 Fama–French sector portfolios and finally Section 7 concludes.

2. The systematic effects of network exposure

2.1. Network models in finance

Network models have featured in an extremely diverse array of applications: in social sciences with studies related to social networking, in natural sciences with application to protein interactions, in government intelligence for analysing terrorist networks, in politics with application to bill coauthorship, in economics with the potential to be used in labour market analysis, and in many other areas. In finance, network models have most frequently been used to assess financial stability as in [Acemoglu et al. \(2012\)](#). In fact, interconnections between financial institutions create potential channels for contagion and the amplification of shocks to the financial system that can also propagate into the “real economy”. Theoretical and empirical studies in this area have garnered considerable interest in the aftermath of the 2007–2009 financial crisis. Network representation of interconnections ranges from linkages extracted from balance-sheet information to connections estimated by means of econometric approaches based on market, accounting, or macroeconomic data.

In order to evaluate the relevance and the price of interconnections in the financial system, it is fundamental to understand all of the channels by which small and local shocks can become big and global.

Empirical network modelling has been conducted to assess asset pricing linkages via contagion ([Allen & Gale, 2000](#); [Bianchi, Billio, Casarin, & Guidolin, 2019](#); [Billio et al., 2012](#); [Dasgupta, 2004](#); [Diebold & Yilmaz, 2014](#); [Hautsch et al., 2015, 2012](#); [Leitner, 2005](#)), linkages via balance sheets ([Cifuentes, Ferrucci, & Shin, 2005](#); [Lagunoff & Schreft, 2001](#)) and how failures of institutions result from mutual claims on each other ([Billio et al., 2014](#); [Furfine, 2003](#); [Ozdagli & Weber, 2017](#); [Upper & Worms, 2004](#); [Wells, 2004](#)). [Allen and A. \(2009\)](#) provide a review of network models in finance.

Much of the empirical finance literature has focused on “direct” contagion arising from firms’ contractual obligations. Direct contagion occurs if one firm’s default on its contractual obligations triggers distress (such as insolvency) at a counterparty firm. Researchers’ simulations using actual interbank loan data suggest that “domino defaults” arising from contractual violations are very unlikely (see [Alves, Ferrari, Franchini, Heam, Jurca, et al. \(2013\)](#), [Degryse and Nguyen \(2004\)](#), [Elsinger, Lehar, and Summer \(2005\)](#), [Furfine \(2003\)](#), [Lelyveld and Liedorp \(2006\)](#), [Mistrulli \(2011\)](#), [Upper and Worms \(2004\)](#)), though they can be highly destructive in the event that they do materialize.

As described above, there are various approaches for estimating networks among financial institutions. However, as far as we know, none of the approaches seems to prevail or to be more appropriate. Our purpose is not to clarify this ambiguity but to show how linkages existing among asset impact returns, asset and portfolio risk, and the pricing of a factor’s risk premia.

Formally, we represent networks by a K -dimensional square matrix W . Each entry $w_{i,j}$ represents the possible connection between assets i and j . A zero entry indicates that the two assets are not connected, while a non-null entry indicates the existence of a connection. We also assume that $w_{i,i} = 0$ for all i , i.e. no asset influences itself, and that the existence of links is not bi-directional, i.e. the network might not be symmetric. Finally, following [Elhorst \(2003\)](#), we normalize W so that, if we are monitoring only the existence of the connection, we equalize the impact of each unit on all other units. We discuss the normalization of W further in a later section.

2.2. A network-augmented linear factor model

Our paper aims at introducing into a multifactor asset pricing model the impact of contemporaneous links that exist across assets, when those links are captured by a network. As discussed in the previous section, networks provide information on the existence of links and might also convey details of the intensity of the links existing between assets. Therefore, we aim to couple the systematic and idiosyncratic risks with a sort of network risk that would introduce into the pricing model the assets’ cross-dependence beyond that captured by common factors. By doing so, we exclude the possibility that the network effects are driven by what [Manski \(1993\)](#) calls *correlated effects*—i.e., common factors. Given this further element, we then evaluate the effects on traditional uses of the multifactor model.

Let us assume that the risky assets are interconnected and that those links can be represented by a network. The network relations, as observed in the previous section, can be in some sense forward looking or represent the actual state of the connections across assets. Following [Manski \(1993\)](#), we could interpret network connections in different ways. They could correspond to correlated network effects associated with missing factors. Alternatively, they might relate to common characteristics, which have not been included in the analysis, but make nodes similar and therefore connected. Furthermore, network effects could stem from the so-called peer effects. We do not have a clear answer to these competing explanations, similarly to the current literature that provides evidence supporting the existence of networks from different viewpoints; see [Cohen, Frazzini, and Malloy \(2008\)](#), [Pool, Stoffman, and Yonker \(2015\)](#), and [Kou et al. \(2017\)](#), among others.

We are tempted to say that various network effects might coexist. Our aim is to extend the asset pricing model in order to take into consideration network effects on top of common factor exposure, as in [Kou et al. \(2017\)](#). Similarly to the latter paper, we allow for network effects and analyse their impact on asset pricing. However, we also introduce a number of additional elements:

(i) a detailed analysis on the role of network exposure on assets' returns and risk; (ii) a discussion on the impact of network effects on the assets' exposure to common risk factors; (iii) the evaluation of network exposures on diversification; (iv) the introduction of asset-specific reactions to networks; (v) the introduction of time changes in the network structure; (vi) the evaluation of (iv) and (v) on risk premium estimation both by proposing an ad hoc estimation approach and evaluating its performance by means of simulation analyses.

From this point onward, we assume that a single network exists, that it is known, that it consists of elements going beyond a set of common factors, and that the network will impact contemporaneous relations across assets. We will show that the presence of interconnections implies that risky assets are exposed to the movements (both systematic and idiosyncratic) of other risky assets. Moreover, we will highlight how risky assets might differ in terms of interconnections with other assets. This creates an additional form of heterogeneity, going beyond those associated with the different exposures to common risk factors and with the different degrees of relevance of idiosyncratic shocks. Starting from this assumption, we reconsider the interpretation of a general multifactor model. If we postulate about the existence of contemporaneous relations across risky assets, we must acknowledge that such relations are not explicitly accounted for in a standard linear factor model of the following form:

$$R_t = \alpha + \beta F_t + \varepsilon_t. \quad (1)$$

where R_t is a K -dimensional vector of time- t risky asset returns, α is a K -dimensional vector of intercepts, β is a $K \times M$ matrix of parameters monitoring the exposure of the risky assets to the common factors included in the M -dimensional vector F_t , and ε_t is the vector of i.i.d. idiosyncratic shocks with zero mean and covariance matrix Σ_ε . In this setting, the beta matrix cannot be directly linked to both the interconnections and the source of *network* heterogeneity across risky assets. One possible way of reconciling the model in Eq. (1) with the network exposure is to interpret the model as a reduced-form model whose reduced-form parameters (the betas and the error covariance) are functions of structural parameters' network exposure and structural factors, such as macrofactors. The latter thus include the *true* exposure to common structural factors, the exposure to other assets due to the interconnections (or network exposure), and the variance of *structural* idiosyncratic shock.

To shed some light on the previous points, we rewrite the model in Eq. (1) as a structural simultaneous equation system (later called a structural model),

$$A(R_t - \mathbb{E}[R_t]) = \tilde{\beta}F_t + \eta_t, \quad (2)$$

where the non-singular matrix A captures the contemporaneous relations across assets and coexists with the common factors, which are here considered as exogenous variables. Moreover, $\mathbb{E}[R_t] = r_f + \beta\Lambda$ in a traditional linear factor model when we take a pricing perspective. In our setting, we will show that the asset expected returns also depend on the network exposure. In Eq. (2), the covariance of η_t , the structural i.i.d. shocks, represents the structural idiosyncratic risk while the parameter matrix A is associated with assets' interconnections, and thus with a network. Further details on the latter aspect will be given in a few paragraphs' time. If we translate the model in Eq. (2) into a reduced form, we have

$$\begin{aligned} R_t &= \mathbb{E}[R_t] + A^{-1}\tilde{\beta}F_t + A^{-1}\eta_t \\ &= \mathbb{E}[R_t] + \beta^*F_t + \varepsilon_t^*, \end{aligned} \quad (3)$$

where β^* is the matrix of reduced-form betas, equal to $\beta^* = A^{-1}\tilde{\beta}$ and ε_t^* is the vector of reduced-form errors.¹ We stress two relevant elements. Firstly, we observe that the reduced-form parameters β^* are *nonlinear* functions of the interconnections between assets (the matrix A) and of the structural exposure to common structural factors (the matrix $\tilde{\beta}$). Secondly, the covariance matrix of the errors ε_t^* is influenced by the presence of interconnections between assets.

We postulate that a network structure exists; consequently, assets are interconnected, and at the same time, the assets depend on a common set of risk factors. If on these assets we estimate the linear factor model in Eq. (1) without taking the network into account, we have, by construction, that the shocks are correlated.² This is a consequence of the fact that the reduced-form betas can be estimated consistently in Eq. (3) anyway by standard linear regression models, but the residuals' covariance is not coherent with the theoretical expectation—that is, idiosyncratic shocks are uncorrelated. Therefore, the empirical evidence of idiosyncratic shock correlation found in the residuals of multifactor models might be due to the exclusion of contemporaneous relations. This is also coherent with the following finding of Ang et al. (2006): idiosyncratic volatility risk is priced in the cross-section of expected stock returns, a regularity that is not subsumed by size, book-to-market, momentum, or liquidity effects. Nevertheless, this phenomenon cannot be explained by exposure to aggregate volatility risk, meaning that the idiosyncratic volatility risk is priced for reason not related to an “exogenous” factor as the aggregate volatility risk. In addition, if we assume that the network links affect the matrix A , and estimate the model in Eq. (3), the residuals' covariance will be a function of the network links.³

So far, our analysis concerns only observable factors. If the common factors are estimated by means of statistical approaches rather than being observed variables, the network exposure, if present, will be totally destroyed. In fact, *statistical* factors are generally estimated from a reduced-form model. Therefore, if we adopt principal component analysis, or fit a latent factor model, it

¹ We include the star as a suffix to distinguish the reduced-form errors implied by our model from the errors of the traditional multifactor model.

² This holds if we assume that A is not diagonal. However, this is an inconsequential restriction as, if A is diagonal, we do not have contemporaneous relations between assets.

³ Assuming normality for the model innovations η_t , such that $\eta_t \sim N(0, \Omega)$, we have that the reduced-form residuals follow the distribution $N(0, A(W)^{-1}\Omega_\eta A(W)^{-1})$, where the reduced residuals are $\varepsilon_t^* = A(W)^{-1}\eta_t$.

might happen that one of the identified factors represents a sort of proxy or biased estimate of the network exposure, with possible further biases on the estimated factor loadings. Such a problem might be overcome by estimating a latent factor model accounting for contemporaneous links across assets. Examples can be found in the studies of Barigozzi and Brownlees (2019) and Bianchi et al. (2019), who show that, if network links are not known, they might be estimated by looking at the covariance of $A^{-1}\eta_t$. However, in such a case, the economic interpretation of network links might be difficult to recover and could be exposed to estimation error. In addition, the common factors we choose in the model must be exogenous.

Our approach aims at reintroducing contemporaneous relations into the multifactor model, thus allowing both the impact of network exposure and the direct exposure to common structural factors to be recovered. Note that the two elements coexist, and network exposure can be seen as an additional *common* risk source going beyond that of common factors. In this respect, we would like to clarify that we are not going to introduce a new exogenous factor based on the network centrality measures in the same fashion of Ahern (2013).⁴ We might even define the exposure to common factors as the exogenous systematic risk exposure, while the network exposure can be labelled as an endogenous systematic risk exposure. The propagation of shocks through the network creates an implicit interdependence between shocks to the stock returns of connected firms. This relation impacts the common factor exposures (already existing and well identified by the literature) and the way to interpret the systematic risk. A possible default of one of the biggest auto-makers would have devastating and widespread effects on their entire supplier–customer network, as each of the car manufacturers were part of an extensive supply chain. Further, we stress that we label the η_t in Eq. (2) as structural idiosyncratic shock to distinguish it from the vector ε_t^* in Eq. (3) that represents the reduced-form idiosyncratic shock.

The simultaneous equation system in Eq. (2) poses serious challenges for the estimation of the matrix A . We overcome this potential problem by resorting to network links. If we postulate the existence of network connections, which we assume are exogenously provided, we can easily recast the network into an adjacency matrix W as mentioned in Section 2.1. The adjacency matrix can be used to impose a structure on the matrix A . Given the matrix W , as extracted from a network, we can easily specify a spatial autoregressive (SAR) model (see Anselin (1988), LeSage and Pace (2009)):⁵

$$R_t - \mathbb{E}[R_t] = \rho W (R_t - \mathbb{E}[R_t]) + \tilde{\beta} F_t + \eta_t, \quad (4)$$

where the (scalar) coefficient ρ captures the response of each asset to the returns of other assets, as weighted by the corresponding row of W . In the spatial econometrics literature, the matrix W is also called a spatial proximity matrix, as it represents *physical* distances between a set of subjects. In the following, we use *spatial proximity matrix* as a synonym for *adjacency matrix*. Moreover, we assume that the error term η_t has a diagonal covariance Matrix—that is, $\mathbb{V}[\eta_t] = \Omega_\eta$ is diagonal. Such an assumption is required for identification purposes, as we will discuss in the model estimation section. If we assume, as we will do in the following, that the matrix W is known, the expected returns are conditional upon the (known) W .⁶

We suggest calling the model in (4) a network-augmented linear factor model. At the single asset level, the model reads as follows:

$$R_{i,t} = \mathbb{E}[R_{i,t}] + \rho \sum_{j=1}^k w_{i,j} (R_{j,t} - \mathbb{E}[R_{j,t}]) + \tilde{\beta}_i F_t + \eta_{i,t}, \quad (5)$$

where $w_{i,i} = 0$, $w_{i,j} \geq 0$, and $\sum_{j=1}^k w_{i,j} = 1$. Taking a financial point of view, the coefficients in the vector $\tilde{\beta}_i$ represent the exposure to the common factors, or *exogenous* exposure, while the coefficient ρ tracks the *endogenous* risk exposure, which is influenced by the network structure and is thus called network exposure. Further insights on the interpretation of the model coefficients will be given in the following subsections.

The model network-augmented linear factor model of Eq. (4) can be rewritten in a more compact form as follows:

$$(I - \rho W) (R_t - \mathbb{E}[R_t]) = \tilde{\beta} F_t + \eta_t, \quad (6)$$

thus highlighting the fact that spatial proximity and the associated SAR model give a structure to the contemporaneous relation matrix, which is now parametrized as

$$A = I - \rho W. \quad (7)$$

The structural model now includes contemporaneous relations, driven by links or connections across assets, systematic components, and asset-specific shocks. In this model, despite the presence of linkages among the companies—for instance, those associated with the supply chain—it might be the case that the linkages have no impact on the contemporaneous relationships among the companies' equity returns. Such a case can be easily identified by a null ρ coefficient. We now elaborate on the relation between returns, risk, networks, and risk factors.

⁴ Since we do not introduce any new exogenous factor we do not apply the corrections proposed by Harvey, Liu, and Zhu (2016) and Feng, Giglio, and Xiu (2020).

⁵ Anselin (1988) calls the model mixed-regressive spatial-autoregressive. We stick here to the simpler acronym adopted in LeSage and Pace (2009).

⁶ To maintain a simplified notation, we do not report the conditioning with respect to W in the returns expectations.

2.3. Returns, networks, and risk factors

The reaction of one asset to common factors and network exposure becomes clearer once we rewrite the model in a reduced-form representation. In this way, we highlight the impact of the network connections included in W on the reduced-form parameters (the reduced-form betas and the reduced-form shock's covariance). The reduced-form representation of the network-augmented linear factor model reads as in Eq. (3) with $A = I - \rho W$ now:

$$\begin{aligned} R_t &= \mathbb{E}[R_t] + (I - \rho W)^{-1} \bar{\beta} F_t + (I - \rho W)^{-1} \eta_t \\ &= \mathbb{E}[R_t] + \beta^* F_t + e_t^*, \end{aligned} \quad (8)$$

where the factor loadings equal $\beta^* = (I - \rho W)^{-1} \bar{\beta}$, and for the moment we assume that $A = I - \rho W$ is nonsingular.⁷ For simplicity, we focus on the case in which the network exposure is driven by a single parameter, ρ . However, all derivations and comments also apply to the more general parametrizations of the matrix A that we will introduce in Section 3.

From LeSage and Pace (2009), we take the following relation:

$$(I - \rho W)^{-1} = I + \rho W + \rho^2 W^2 + \rho^3 W^3 \dots, \quad (9)$$

where the term ρW monitors the effect of linked assets (in spatial econometrics, the neighbours). For instance, if asset j is linked to asset i we have a non-null entry in W_{ij} . Differently, $\rho^2 W^2$ is associated with the effect on asset j induced by the assets linked to asset i (in spatial econometrics, the second-order neighbours). The latter relation can be further generalized to higher orders. Notably, the matrices W^j might also include a so-called *feedback loop* as, following the previous example, asset i can be linked to asset j (making the relation bidirectional), causing the matrix W^j to have non-null elements on the main diagonal. We stress that, despite the fact that the summation has an infinite number of terms, by imposing that $|\rho| < 1$, we can easily ensure that the effect of linked assets converges to a finite number. On the contrary, if $|\rho| > 1$, we might have explosive patterns. In general, the coefficient ρ takes values in the range $(\lambda_{min}^{-1}, \lambda_{max}^{-1})$, with λ_{min} and λ_{max} , respectively, the minimum and maximum eigenvalues of W . In the case of the W matrix's row normalization, in spatial econometrics, a commonly adopted range is $[0, 1)$.

Using Eq. (9), we can rearrange the network-augmented linear factor model of Eq. (8) as

$$R_t = \mathbb{E}[R_t] + \bar{\beta} F_t + \sum_{j=1}^{\infty} \rho^j W^j \bar{\beta} F_t + \eta_t + \sum_{j=1}^{\infty} \rho^j W^j \eta_t. \quad (10)$$

Such a representation highlights that the impact of the common factors, as well as that of the idiosyncratic shocks on the risky asset returns, can be decomposed into two parts. For both elements, the first component is the traditional, direct, or structural impact of the structural common factor, while the second component is the impact associated with network exposure. We can thus define the following four elements:

- (a) $\bar{\beta}$: the structural exposure to the structural common factor;
- (b) $\sum_{j=1}^{\infty} \rho^j W^j \bar{\beta}$: the network exposure to the structural common factor;
- (c) η_t : the idiosyncratic shocks;
- (d) $\sum_{j=1}^{\infty} \rho^j W^j$: the network impact of idiosyncratic shocks.

Note that the network-related exposures depend on the structure of the matrix W as well as on the parameter monitoring the network impact, ρ . A relevant remark can be made in relation to the network impact of common factors. Let us take, for simplicity, a specific common factor. That is, we focus on a single column of F_t and consider the impact of the m th factor on the risky asset returns:

$$\beta_m^* = \bar{\beta}_m + \sum_{j=1}^{\infty} \rho^j W^j \bar{\beta}_m. \quad (11)$$

Starting from the observation that the reduced-form betas equal the sum of two elements, Eq. (11) provides two relevant insights.

First, we note that the network exposure to common factors acts as a multiplier of the structural exposure if the coefficient ρ is positive (W 's elements are positive anyway). Therefore, shocks to the common factors will be amplified by (i) the presence of connections across assets—that is, when, for asset i , the i th row of W contains at least one non-null element, (ii) the change in the impact of network connections—that is, when the coefficient ρ increases, and (iii) by changes in the network structure—that is, when the matrix W changes. Note that, if asset i is not connected to other assets, all products $\rho^j W^j \bar{\beta}_i$ are equal to zero.

From a different viewpoint, the presence of network exposure allows us to decompose the reduced-form betas into two components, a structural one and a multiplier depending on the network structure, the matrix W . The estimation of a standard factor model, where the data-generating process includes network dependence across returns, will provide partial information, returning only the combination of the two components—that is, only the reduced-form betas. Furthermore, by using this model, we can disentangle the presence of network-effects-induced exposure to common factors from the structural exposure to common factors. If network effects do not play a role, we expect the ρ coefficient to be zero—that is, the network among the companies has no impact on the contemporaneous relations between the companies' equity returns.

⁷ See the model estimation section for details on the constraints that ensure the nonsingularity of A .

Now assume that, for the risky asset i , the m th common factor is not relevant (that is, $\bar{\beta}_{i,m} = 0$). In this case, in the standard linear factor models, the common factor will have no role in explaining the asset returns. However, when assets are linked and network exposures are taken into account, a common factor to which a risky asset has zero structural exposure might still be relevant for explaining the evolution of the risky asset return. Such an effect is not direct, but induced from the network exposure, and is associated with the existence of non-null elements in the i th row of the matrix W . Take, for instance, the following case:

$$W = \begin{bmatrix} & \vdots & & \\ \mathbf{0}_i & 1 & \mathbf{0}_{K-i-1} & \\ & \vdots & & \end{bmatrix}, \quad (12)$$

where asset i is connected only to asset $i + 1$ and the subscripts denote the lengths of the row vectors of zeros. Moreover, assume the following factor exposure for both assets:

$$\bar{\beta} = \begin{bmatrix} & \vdots & & \\ \bar{\beta}_{1,i} & 0 & 0 & 0 \\ \bar{\beta}_{1,i+1} & \bar{\beta}_{2,i+1} & 0 & 0 \\ & \vdots & & \end{bmatrix}, \quad (13)$$

where, in a multifactor model, asset i is not exposed to factor 2, while asset $i + 1$ is affected by the same risk factor, and both assets are exposed to factor 1. Asset i 's dependence on risk factors can thus be represented as

$$\bar{\beta}_{1,i} F_{1,t} + \rho \bar{\beta}_{1,i+1} F_{1,t} + \rho \bar{\beta}_{2,i+1} F_{2,t} + \sum_{j=2}^{\infty} (\rho^j W^j \bar{\beta} F_1) |_i, \quad (14)$$

where $|_i$ identifies the i th element of a vector. Note that Eq. (14) shows that, because of the network effect, asset i increases the exposure to factor 1 of $\rho \bar{\beta}_{1,i+1}$ and is indirectly exposed to factor 2 of $\rho \bar{\beta}_{2,i+1}$, and that the last term on the right represents further elements that can be specified only through the knowledge of the entire matrix W . Therefore, even if risky asset i is not (structurally) exposed to a common factor (in the previous example, factor 2), the common factor will still play a role if it impacts the returns of the assets to which i is linked.

Such a result can be further generalized by focusing, for instance, on sector-specific risk factors. Those factors, in the presence of a network exposure, despite being sector specific, will have a systematic impact on all connected assets. Moreover, if we disregard the network exposure, we might also incur the risk of misinterpreting the impact of risk factors. In fact, by estimating the reduced-form model, we might label as *common* a factor that in reality is structurally related just to a subset of the investment universe and that impacts other assets only through network connections.

A similar property exists for the idiosyncratic shocks. In fact, if we assume they are uncorrelated, the existence of network connections implies that the structural shocks of one asset impact on the returns of all the assets connected to it. Therefore, shocks on single assets can have effects on many other risky assets.⁸ This evidence on the impact of a given asset's factor exposures and the shocks of linked assets resembles the decompositions typically adopted in spatial econometrics; see, among many others, [LeSage and Pace \(2009\)](#), as well as [Ozdagli and Weber \(2017\)](#) for a recent application.⁹ We note that the latter decompositions are appropriate in a framework where, for given dependent-variable measures across subjects, we have a number of covariates, each of which is available with variable-specific observations. In our case, we do have common factors (not asset-specific variables) and therefore these decompositions cannot be applied.

From a pricing perspective, starting from the reduced-form representation of the network-augmented linear factor model, it is possible to show that the expected returns equal

$$\mathbb{E}[R_i] = r_f + \bar{\beta} \Lambda + \sum_{j=1}^{\infty} \rho^j W^j \bar{\beta} \Lambda, \quad (15)$$

see [Kou et al. \(2017\)](#) for a formal proof. Expected returns are thus influenced by network links that amplify the compensation for being exposed to the common factors. Further, we note that the pricing result depends on, and is thus conditional upon, the network structure, as summarized by W , which we assume to be known and time invariant. In fact, if we postulate that the coefficient ρ is positive and that the elements of W are all positive, the existence of links across assets induces higher expected returns than in the case of links being absent. Moreover, bearing in mind the previous discussion, the expected returns might depend on risk premia associated with factors to which a given asset is not directly (structurally) exposed. The model estimation section discusses further elaborations on the risk premia that emerge as consequences of the risk premium estimation.

In addition, we stress that the use of a network that is very dense, thus implying a matrix W that is almost full, will have further impacts. In fact, a full W implies that all idiosyncratic shocks are correlated. However, from our viewpoint, this corresponds

⁸ Summary measures of the exposure to common factors and idiosyncratic shocks can be obtained by mimicking the approaches used in spatial econometrics. A discussion on this topic is included in [LeSage and Pace \(2009\)](#); see their Section 2.7. These measures have been used in a financial framework by [Asgharian, Hess, and Liu \(2013\)](#). We also note that the decomposition of asset returns into four elements is equivalent to that proposed by [Abreu, De Groot, and Florax \(2005\)](#) for separating the standard impact of covariates from the impact that is due to the spatial links, and is thus an alternative to the impact measures of [LeSage and Pace \(2009\)](#).

⁹ The latter use spatial auto-regressions to decompose the overall effect of monetary policy shocks into a direct effect and an indirect (network) effect. They attribute 50%–85% of the overall effect to indirect effects.

to indirect evidence of model misspecification as an additional common factor is now present but not taken into account. As a consequence, such common structural factor risk must be priced, and could generate the empirical evidence shown by Ang et al. (2006). The latter case could also correspond to empirical evidence challenging the validity of the APT approach. From a different viewpoint, our modelling framework still satisfies the assumptions required for APT. As we show in the next section, the presence of a network exposure, despite inducing correlation across the idiosyncratic shocks, does not exclude the existence of diversification benefits. In line with the network literature, we can also establish a link between our model and a measure of network centrality. In fact, Katz (1953) uses the expression in Eq. (9) to introduce his centrality measures. If we denote by $\mathbf{x}$ the score vector of the centrality, while W is the adjacency matrix, $\mathbf{1}$ a vector of ones, and α and C arbitrary constants, then Katz (1953) shows that the centrality vector equals $\mathbf{x} = (I - \psi W)^{-1} \mathbf{1}$, with ψ being a free parameter. Note that the ψ manages the relationship between the centrality vector $\mathbf{x}$ and a limiting constant value for all centrality scores. In other words, if ψ is zero, all the nodes have the same centrality value. Otherwise, if ψ assumes increasing values,¹⁰ then W plays an increasing role.

2.4. Risk decomposition and portfolio diversification

The network-augmented linear factor model of Eq. (2) allows us to recover a risk decomposition similar to that available for the standard linear factor model in Eq. (1). The starting point is the reduced form introduced in Eq. (8). In fact, Eq. (8) highlights that the estimation output of standard multifactor models can be consistent with the presence of contemporaneous links across assets, and this allows us to elaborate on the returns covariance structure. In fact, we can redefine $\mathcal{A} = (I - \rho W)^{-1}$, and then write the total variance of the risky assets as follows:

$$\mathbb{V}[R_t] = \mathcal{A} \bar{\beta} \bar{\beta}' \mathcal{A}' \sigma_m^2 + \mathcal{A} \Omega_\eta \mathcal{A}'. \quad (16)$$

Despite being equivalent to the traditional risk decomposition of a multifactor model, Eq. (16) provides a relevant insight. In fact, both the systematic and idiosyncratic risk components are influenced by the presence of interconnections across assets, as the matrix $\mathcal{A}$ appears in both terms on the right-hand side. This also shows that, if we estimate the reduced-form model using standard linear methods, our evaluations of the systematic and idiosyncratic risk components are in reality a blend between the structural loadings, the idiosyncratic risks, and the network relations. Eder and Keiler (2015) suggest that the presence of spatial links could be interpreted as a systemic risk contribution. They interpret it as a contagion effect within a financial system. However, the previous decomposition provides an alternative view, in which spatial dependence is not an additive source of risk but rather a multiplicative one. In that case, the impact of the spatial dependence of a single asset cannot easily be recovered. In fact, the contribution to the asset risk due to the spatial dependence depends on both the structure of the network W and the spatial parameter ρ . The two elements impact on the systematic contribution and on the idiosyncratic structural shock variances in a nonlinear way. Obviously, the same structure appears at the portfolio level, where we have

$$\mathbb{V}[r_{p,t}] = \omega' \mathcal{A} \bar{\beta} \Sigma_F \bar{\beta}' \mathcal{A}' \omega + \omega' \mathcal{A} \Omega_\eta \mathcal{A}' \omega. \quad (17)$$

We first note that, if asset interconnections are not present (that is, when $\mathcal{A} = I$), the idiosyncratic risk equals Ω_η while the systematic risk component is $\bar{\beta} \Sigma_F \bar{\beta}'$. We rewrite the portfolio variance decomposition in Eq. (17) by adding and subtracting the portfolio idiosyncratic and systematic variance components when they are not influenced by asset interconnections:

$$\mathbb{V}[r_{p,t}] = \omega' \mathcal{A} \bar{\beta} \Sigma_F \bar{\beta}' \mathcal{A}' \omega + \omega' \mathcal{A} \Omega_\eta \mathcal{A}' \omega \pm \omega' \bar{\beta} \Sigma_F \bar{\beta}' \omega \pm \omega' \Omega_\eta \omega. \quad (18)$$

With some rearrangement, the total portfolio variance can be recast into a decomposition comprising four different terms:

$$\begin{aligned} \mathbb{V}[r_{p,t}] = & \underbrace{\omega' \bar{\beta} \Sigma_F \bar{\beta}' \omega}_I + \underbrace{(\omega' \mathcal{A} \bar{\beta} \Sigma_F \bar{\beta}' \mathcal{A}' \omega - \omega' \bar{\beta} \Sigma_F \bar{\beta}' \omega)}_{II} \\ & + \underbrace{\omega' \Omega_\eta \omega}_{III} + \underbrace{(\omega' \mathcal{A} \Omega_\eta \mathcal{A}' \omega - \omega' \Omega_\eta \omega)}_{IV}. \end{aligned} \quad (19)$$

We make the following interpretations of the four risk components derived from our network-augmented linear factor model:

- I the structural systematic risk component that depends on the structural loadings from the common factors and from the covariance of the common factors; this is the *exogenous* systematic effect;
- II the effect of asset interconnections on the systematic risk component, or the first contribution of network exposure to the total risk; this is the *endogenous* systematic effect;
- III the structural idiosyncratic component that depends only on the covariance of the structural shocks;
- IV the effect of interconnections on the idiosyncratic risk, or the second contribution of network exposure to the total risk; this might be interpreted as an *endogenous* amplification of idiosyncratic risks.

¹⁰ With an upper limit at $1/\lambda_{\max}$, $\lambda_{\max}$ being the maximum eigenvalue of W , the value for which $(I - \psi W)^{-1}$ is nonsingular.

Note that by adding the second and fourth terms, we obtain the total contribution of network exposure to the total portfolio risk. We finally note that the model with asset interconnections is the standard multifactor model if there are no interconnections—that is, if W is a null matrix or if the coefficient ρ is zero.

In addition, the network exposure impacts the idiosyncratic part of the variance. This implies that the diversification benefits might be endangered, depending on the network structure. In fact, despite the fact that the fourth term decreases with an increase in these cross-sectional dimensions, the speed of decrease is smaller than in the case without network effects.

Similar to the standard linear factor model, we can recover analytical elements in a simplified setting. The covariance matrix Ω_η is diagonal; we further assume that the diagonal elements are set to an average value, $\bar{\sigma}^2 = 1$. In addition, we take an equally weighted portfolio and focus on the limiting case in which all assets are connected (thus W has zeros only over the main diagonal, while off-diagonal terms equal $\frac{1}{K-1}$ after row normalization). In this case, we have that

$$\begin{aligned}\omega' \mathcal{A} \Omega_\eta \mathcal{A}' \omega &= \bar{\sigma}^2 \omega' \mathcal{A} \mathcal{A}' \omega = \frac{\bar{\sigma}^2}{K^2} \mathbf{i}'_K \mathcal{A} \mathcal{A}' \mathbf{i}_K \\ &= \frac{K + \rho^2}{(K + \rho)^2 (\rho - 1)^2} \bar{\sigma}^2,\end{aligned}\quad (20)$$

where K is the number of assets and $\mathbf{i}_K$ is a K -dimensional vector of ones.¹¹ Moreover, we have that

$$\lim_{K \rightarrow \infty} \frac{K + \rho^2}{(K + \rho)^2 (\rho - 1)^2} \bar{\sigma}^2 = 0, \quad (21)$$

thus preserving the diversification benefit. However, the idiosyncratic risk contribution is higher than in the case without spatial dependence (i.e., with $\rho = 0$). In fact, we can show that the above-reported portfolio idiosyncratic risk is higher than $\frac{1}{K} \bar{\sigma}^2$, thus confirming that term IV is positive.

The previous model thus gives a framework in which we can analyse the impact at the portfolio level of the interconnections we might observe across assets, and how those interconnections can endanger/limit the benefits of portfolio diversification.

3. Model generalizations

In this section, we provide two generalizations of the network-augmented linear factor model by allowing asset-specific reaction to the network and introducing a time change into the network structure. We also add further discussion on the model's interpretation, in particular on the sign of the network-related parameters.

3.1. Heterogeneous network reaction

The model in Eq. (6), despite being coherent with the presence of network links, has a very restricted structure. There is a single parameter, ρ , driving the network exposure. This can easily be generalized by allowing asset-specific responses to the network structure. We can thus modify the contemporaneous relation matrix of Eq. (7) into

$$A = I - RW, \quad (22)$$

where $R = \text{diag}(\rho_1, \rho_2, \dots, \rho_K)$ is a diagonal matrix. This model is similar to the fixed coefficient specifications for spatial panels discussed in Elhorst (2003). A clear advantage of such a structure is given by the possibility that assets have different network exposures, as for each asset the model becomes

$$R_{i,t} = \mathbb{E}[R_{i,t}] + \rho_i \sum_{j=1}^k w_{i,j} (R_{j,t} - \mathbb{E}[R_{j,t}]) + \bar{\beta}_i F_t + \eta_{i,t}. \quad (23)$$

To estimate the asset-specific parameters, the network must satisfy an identification condition: each asset must be connected to at least one other asset. If this is not the case, the diagonal of matrix R must be restricted in such a way that unconnected assets will not have a network exposure. Further details will be discussed in the estimation section.

3.2. On the sign of the coefficient ρ

Up to this point, we have not discussed the sign of the coefficient ρ . Intuitively, we expect the assets to be positively related to one another, as the links come from a network. We thus imagine that shocks are transmitted to connected assets with their signs preserved. If we take a simplified model with a single coefficient ρ , it is highly improbable that we will ever observe negative coefficients. In fact, a single coefficient represents a sort of average reaction of the asset to the shocks coming from its neighbours.

However, in a model accounting for the heterogeneity of the reaction to the network exposure, negative asset-specific coefficients might appear. In other words, we cannot exclude a priori that a shock to one asset will lead to an opposite movement of a linked

¹¹ In the special case considered, the diagonal elements of $\mathcal{A}$ equal $\frac{(K-1)\rho-K}{\rho^2+(K-1)\rho-K}$ and the off-diagonal elements are $\frac{-\rho}{\rho^2+(K-1)\rho-K}$. Moreover, the diagonal elements of $\mathcal{A}\mathcal{A}$ equal $\frac{K\rho^2+[(K-1)\rho-K]^2}{[\rho^2+(K-1)\rho-K]^2}$ and the off-diagonal are $\frac{(K-1)\rho^2-2\rho[(K-1)\rho-K]}{[\rho^2+(K-1)\rho-K]^2}$. Summing up the elements in $\mathcal{A}\mathcal{A}$ and simplifying, we obtain the above-reported result.

asset. We explain such a finding by making a parallel with negative correlations. If two assets are negatively correlated, their joint introduction to a portfolio will lead to a decrease of the overall variance. In a factor model, negative correlations across asset returns can be interpreted by loadings on the (same) common factors having different signs. In our framework, negative correlations across asset returns can emerge both in response to different signs in the factor loadings and also due to the presence of a negative asset-specific reaction to the network exposure.

Consider the reduced form of our model as represented in Eq. (19). In this case, the innovation term has a nondiagonal covariance. If we estimate the reduced-form model, the innovations could show evidence of non-null correlations, some of them being negative. They can be due to the presence of opposite exposure to the common structural factors, whose coefficients have been estimated by a biased estimator (due to model misspecification), but can also be due to the presence of negative coefficients ρ_i .

In a general model with heterogeneous asset reactions to the network exposure, the components II and IV in the risk decomposition presented in Eq. (19) can become negative. In such a case, the network exposure reduces risk, and this could also be seen as a kind of flight-to-safety effect: if shocks hit financial assets and are then transmitted to industrial procyclical sectors, we cannot exclude the possibility that the anticyclical sectors have an opposite network exposure.

Within our model, negative ρ might thus exist, but how can we interpret them from a pricing perspective? We read them as evidence of risk absorption due to network exposure. In fact, a negative ρ_i allows a reduction of the exposure of one asset to the common factors, since the i th component of the second term in Eq. (11) becomes negative. Risk absorption also has consequences for expected returns, leading to a reduction of the contribution of network exposure. In fact, the i th component of the third term in Eq. (15) also becomes negative.

3.3. Time change in the network structure

The spatial econometrics literature generally assumes that the spatial proximity matrix is time invariant. In fact, if the matrix W depends on physical measures, such as those is the spatial distance, they can safely be assumed to be constant over time. However, in a financial framework, the connections between assets might change over time for a number of reasons, such as, for instance, the occurrence of an unexpected market shock, or a merger or acquisition. Similar approaches have been adopted by Asgharian et al. (2013) and Eder and Keiler (2015). We are still assuming that the network is exogenous with respect to the linear structural model,¹² and the contemporaneous matrix can be further rewritten as

$$A_t = I - \mathcal{R}W_t, \quad (24)$$

where we highlight that the network changes over time, and thus leads to a time-varying matrix W . In turn, this induces time-dependence on the matrix A , as well as on the reduced-form parameter matrices, both on the betas and on the covariance of idiosyncratic shocks. That is, we also have heteroskedasticity. Nevertheless, we might postulate that the dynamic of W_t is smooth and operates at lower time scales than those monitoring the evolution of returns (for instance, we can assume the matrices W change over the years, or after specific events such as crises). Therefore, the heteroskedasticity is mild, and the betas are evolving slowly. The use of time-varying matrices W thus leads to a time change in the spatial dependence, differing from the approach of Blasques, Koopman, Lucas, and Schaumburg (2016), who obtain the same result by letting the parameters $\mathcal{R}$ be time varying. We note that, if the network exposure exists and the structural parameters in the matrix $\tilde{\beta}$ are constant, the estimation of the reduced-form model over different samples might suggest changes in the factor exposure. However, those changes are not present, but are due to the misspecification of the network relations. The expected returns are conditional upon the matrix W . If the network exposure is time varying, the expected returns, conditional on W_t , are also time varying.

A further issue associated with the change in W_t over time is the normalization. In fact, if we let each single W_t be row normalized, we could reduce the impact of changes in the network density: an increase in the number of assets linked to asset j would lead to a decrease in the impact arising from a single asset, since the corresponding element of W_t would diminish. As a consequence, with the introduction of the dynamic W_t , we also suggest considering a different normalization, which we refer to as a *max row normalization*. Formally, a non-normalized W_t^U will be normalized as

$$W_{i,j,t} = W_{i,j,t}^U \left(\max_i \sum_{j=1}^N W_{i,j,t}^U \right)^{-1}. \quad (25)$$

We stress that, when conditioning on the network structure, the pricing equation changes with respect to that in Kou et al. (2017).¹³ In fact, the pricing equation, for the most general network-augmented linear factor model, assumes the following form (in which we have also introduced asset-specific coefficients for network exposure):

$$\mathbb{E}[R_t | W_t] = r_f + \tilde{\beta}\Lambda + \sum_{j=1}^{\infty} (\mathcal{R}W_t)^j \tilde{\beta}\Lambda. \quad (26)$$

The heterogeneity with respect to connections creates reactions to shocks on the common factors that differ across assets due to the different exposures of assets to the factors, but also due to the differing impacts of feedback loops coming from the underlying

¹² We might relax the exogeneity assumption by stating that the network is a known conditional upon the past.

¹³ In Kou et al. (2017) W is time-invariant.

network structure. The change over time of the matrix W_t , or the presence of a structural break on the coefficients $\mathcal{R}$ (that we might find located close to the time of a crisis or extreme event), creates abrupt changes in the expected returns, and relevant movements in stock prices as a consequence. Therefore, the pricing, conditional upon the network structure, becomes a function of the network structure: if the network changes, the *local equilibrium* expected returns change. When we introduce a time variation into the matrices W , or into the elements $\mathcal{R}$, we must estimate risk premia in the cross-sectional dimension. However, deviating from standard approaches, we should use reduced-form model parameters and account for their time variation, as we will show in Section 4.1. We stress that if we focus on a standard pricing model, we neglect the potential time variation in the expected returns, with a possible impact on the estimation of the risk premia. As an alternative approach, we could estimate the risk premium by taking the evolution of the network structure into account. However, this requires an assumption of the stochastic process that governs the network evolution. The latter is well beyond the scope of this paper and is left to upcoming research. In Section 6, we focus on a special case for the network evolution, and we show how correctly accounting for the time variation of the network structure provide an improvement in the efficiency of risk premium estimators.

4. Model estimation

In Sections 3 and 4, we formally introduced the theoretical framework of this paper, described the network-augmented linear factor model, and provided a number of model generalizations. In this section, we first discuss the approach we propose to estimate model parameters-i.e., the factor structural loadings $\bar{\beta}$, the network impact parameters ρ_i , $i = 1, 2, \dots, k$ included in the matrix $\mathcal{R}$, the error covariance Ω_η , and the expected returns. Second, we move to the estimation of the risk premia following a two-step approach, which is conditionally to the availability of the estimates of the model parameters. Both approaches pose various challenges that are related to the model structure, in particular from the existence of contemporaneous dependence driven by the network.

For estimating parameters, let us consider the simultaneous equations model

$$AR_t = \alpha + \bar{\beta}F_t + \eta_t. \quad (27)$$

Identification conditions are required to estimate the parameters of A , α , $\bar{\beta}$, and the (diagonal) covariance matrix of η_t . The simple order condition of identification requires that the model parameters must be less than the parameters we can recover from the reduced-form specification. In fact, the latter can be estimated by least squares methods, and the structural parameters can be recovered thanks to their relation to the reduced-form parameters. The reduced-form model is

$$\begin{aligned} R_t &= A^{-1}\alpha + A^{-1}\bar{\beta}F_t + A^{-1}\eta_t \\ &= \alpha^* + \beta^*F_t + \varepsilon_t^*, \end{aligned} \quad (28)$$

suggesting we can consistently estimate $K(M+1)$ mean parameters plus $\frac{1}{2}K(K+1)$ covariance parameters, where M is the number of factors and K the number of assets. However, an unrestricted structural specification, despite having the same number of parameters in the covariance, has $MK + K$ mean parameters.

The presence of asset interconnections, summarized into a network, allows a sensible reduction of the number of parameters included in the matrix A . In fact, if we have asset-specific network exposures and a single network, we have only K parameters in A . However, this is not sufficient to provide identification of the model's remaining parameters, since the order condition is still not satisfied. Identification is obtained by imposing the diagonality of the covariance matrix of η_t . Such a choice, which is economically motivated, allows the standard order condition for identification to be satisfied.

Nevertheless, further constraints on the model parameters are generally required. Starting from the spatial econometrics literature, which takes a scalar time-invariant coefficient ρ and a time-invariant row-normalized matrix W , we must impose $\frac{1}{\lambda_{\min}} < \rho < \frac{1}{\lambda_{\max}}$, where $\lambda_{\min}$ and $\lambda_{\max}$ are, respectively, the minimum and maximum eigenvalues of W . This constraint ensures the nonsingularity of $A = I - \rho W$.

In our framework, we deviate from traditional approaches in several ways. We first consider the case of a time-varying adjacency matrix-that is, W_t . A sufficient condition for the invertibility of $I - \rho W_t$ for all t is stated in the following assumption:

Assumption 4.1. The coefficient ρ satisfies the following condition:

$$\bar{\lambda}_{\min}^{-1} < \rho < \bar{\lambda}_{\max}^{-1}, \quad (29)$$

where

$$\bar{\lambda}_{\max} = \min \{ \lambda_{t,\max} \}_{t=1}^T \quad (30)$$

$$\bar{\lambda}_{\min} = \max \{ \lambda_{t,\min} \}_{t=1}^T \quad (31)$$

and $\lambda_{t,\max}$ and $\lambda_{t,\min}$ are, respectively, the minimum and maximum eigenvalues of a matrix W_t .

If we have a diagonal matrix $\mathcal{R}$ containing the asset-specific reaction to the spatial links, we assume the nonsingularity, which is then validated in the estimation step of the model:

Assumption 4.2. The diagonal coefficient matrix $\mathcal{R}$ is such that

$$I - \mathcal{R}W_t \quad (32)$$

is nonsingular for each matrix W_t .

Note that the previous assumption covers both the case of a time-invariant and a time-varying adjacency matrix. We further note that, when we consider a model with $\mathcal{R}$, we must impose an additional identification condition:

Assumption 4.3. The diagonal coefficient matrix $\mathcal{R} = \text{diag}(\rho_1, \rho_2, \dots, \rho_K)$ is such that $\rho_j = 0$ if the matrix $W_j = [W'_{j,1} \ W'_{j,2} \ \dots \ W'_{j,T}]$, with $W_{j,t}$ being the j th row of W_t , has non-null rank.

The previous assumption requires, irrespective of the number of matrices W_t , that if the j th rows of all the matrices W_t contain only zeros (that is, the asset j is not linked to any other asset in the varying evolution of the network), then the asset j 's network impact coefficient is restricted to zero as it cannot be identified. This condition ensures that the asset-specific impact on the network links is estimated only if such links exist for at least one point in time.

The use of covariance restrictions has a consequence for the estimation of the model parameters. In fact, they must be jointly evaluated, despite the fact that the linear model structure might allow for single-equation (single-asset) parameter estimation.

Under the two strong parametric restrictions we impose (the structure on A and the absence of correlation across the idiosyncratic shocks), a viable approach is the Full Information Maximum Likelihood (FIML) method. However, if K is even moderately large, the total number of parameters to be estimated in the restricted structural model, MK , might be quite large. Fortunately, we can follow the approaches commonly used in spatial econometrics, namely the use of concentrated likelihoods. As in [Elhorst \(2003\)](#) and [LeSage and Pace \(2009\)](#), we start by writing the full model's log likelihood assuming Gaussianity:¹⁴

$$L(\Theta) = \sum_{j=1}^T l_t(\Theta), \quad (33)$$

$$l_t(\Theta) \propto -\frac{1}{2} \log |\Omega| - \frac{1}{2} e'_t \Omega^{-1} e_t, \quad (34)$$

$$e_t = R_t - \bar{\alpha} - \mathcal{R}W_t R_t - \bar{\beta} F_t, \quad (35)$$

where Ω is a diagonal matrix. If the parameters in $\mathcal{R}$ are known, we can write

$$R_t - \mathcal{R}W_t R_t = Z_t = \bar{\alpha} + \bar{\beta} F_t + \varepsilon_t. \quad (36)$$

Therefore, with a known network exposure parameter matrix $\mathcal{R}$, with a unique (and even time-variant) network structure, we can estimate the parameters in $\bar{\alpha}$ and in $\bar{\beta}$ by least squares methods, obtaining the well-known expressions. In addition, we can even recover standard estimators for the innovation variance. This suggests that the network exposure parameters can easily be obtained by maximizing the concentrated likelihood obtained by replacing the other parameters with their least squares estimators.¹⁵

We thus define $Z = [Z'_1, Z'_2, \dots, Z'_T]$, $X_t = [1, F_t]$, $X = [X'_1, X'_2, \dots, X'_T]$, and the least squares estimators of $\bar{\alpha}$ and $\bar{\beta}$ from (36) equal

$$\begin{bmatrix} \hat{\bar{\alpha}}(\mathcal{R}) \\ \hat{\bar{\beta}}(\mathcal{R}) \end{bmatrix} = (X'X)^{-1} X'Z \quad (37)$$

highlighting that the least squares estimates are functions of the network-related parameters included in $\mathcal{R}$. Similarly, the variance parameters of ε_t correspond to the variance of $\hat{\varepsilon}_t(\mathcal{R}) = Z_t - \hat{\bar{\alpha}}(\mathcal{R}) - \hat{\bar{\beta}}(\mathcal{R}) F_t$. If we use those least-squares estimators to replace the corresponding parameters within the likelihood function, we obtain a concentrated likelihood function depending only on the network-related parameters.

Using concentrated likelihood allows us, at no cost, to deal with zero restrictions added to specific parameters of $\mathcal{R}$, and to easily ensure model identifications. Furthermore, the estimation approach outlined above is also valid when a time-varying adjacency matrix W_t is given. In fact, we postulate that the sequence of W_t is known. Therefore, we estimate the model parameters conditional to the sequence of adjacency matrices. In that case, we just have to define Z_t as

$$Z_t = R_t - \mathcal{R}W_t R_t. \quad (38)$$

Note that the only difference between (36) and (38) is in the time-varying nature of the network. Further, we point out that our framework accounts for cross-sectional dependence among the variables through the network-i.e., the time-varying adjacency matrices W_t . However, we cannot exclude a priori that the networks might be affected by measurement errors. In fact, from an economic viewpoint, the networks might be estimated rather than observed. In a cross-sectional framework, [Conley \(1999\)](#) discusses

¹⁴ We use a Normality hypothesis only for simplicity. More general distributions might be used or, alternatively, we might resort to Quasi-Maximum-Likelihood approaches, which are based on a Normal likelihood.

¹⁵ This is of relevant computational importance as it allows us to reduce the parameters to be jointly estimated to $2K$ if we concentrate the likelihood with respect to $\bar{\alpha}$ and $\bar{\beta}$, and to K if we also concentrate with respect to the innovation variance. Standard errors can be recovered from the full-model likelihood by making numerical evaluations of the Hessian (and of the gradient if we take a robust-parameters covariance matrix).

the estimation of a linear model where subjects are cross-sectionally correlated by means of an economic notion of distance and the distance might be measured with error. His work focuses on the estimation of the covariance matrix, which is contaminated by such an economically driven spatial correlation. We differ from his approach in both the presence of a dynamic network as well as for the specific heterogeneous impact that the network has over the asset returns. In the simulation section, we have a closer look at the evaluation of the effects due to the time-varying networks on the asymptotics of the network effects measured by the parameters included in $\mathcal{R}$.

4.1. Risk premium estimation

We now tackle the crucial issue of risk premium estimation in our framework. The literature generally follows two approaches, the Fama and French (1993) or the Fama and MacBeth (1973) method. Similar to Ahern (2013), for the network factors, we follow the Fama and MacBeth (1973) two-pass method. Therefore, we plan to estimate the risk premia starting from the time-series estimate of the betas and then taking a cross-sectional regression. We thus stress that the risk premia estimations comes after the estimation of the model parameters.

To highlight the links existing between the traditional multifactor model and our proposal, we first rewrite the two models, the linear factor model and the factor model augmented with network-driven contemporaneous relations, respectively, reporting the second in its reduced-form representation:

$$R_t = \alpha + \beta F_t + \varepsilon_t \quad (39)$$

$$R_t = \alpha^* + \beta^* F_t + \varepsilon_t^*. \quad (40)$$

Under the traditional factor model, in which the M factors do have a zero mean, the Fama-McBeth procedure corresponds to a collection of K time-series regressions of the form

$$R_{i,t} = \alpha_i + \beta_i F_t + \varepsilon_{i,t}, \quad (41)$$

followed by a cross-sectional regression

$$R^e = \hat{\beta} \Lambda + v, \quad (42)$$

where R^e is the K -dimensional vector of the sample averages of excess returns, $\hat{\beta}$ is the K -by- M matrix of (least squares) factor-loadings estimates and Λ is the unknown vector of risk premia. The estimates suffer from the error-in-variable issue. In the empirical part, we manage the error-in-variable issue by working on industry portfolios. Further, we introduce the Shanken (1992) correction into the estimation of the standard errors.

We start by focusing on the case with heterogeneous network impact and time-invariant W . In that case, the risk premia and the expected asset returns satisfy the following equilibrium condition:

$$\mathbb{E}[R_t] = r_f + \bar{\beta} \Lambda + \sum_{j=1}^{\infty} \mathcal{R}^j W^j \bar{\beta} \Lambda = r_f + \beta^* \Lambda. \quad (43)$$

Therefore, in our model, the estimation of the risk premium using the Fama-McBeth approach points at using the reduced-form factor loadings β^* . Consequently, we first estimate the model and then run the cross-sectional regression

$$\bar{R}^e = \hat{\beta}^* \Lambda + v^*, \quad (44)$$

where we have replaced the reduced-form betas with their estimates. By simple comparison of the two second-stage (cross-sectional) regressions, we can state the following proposition:

Proposition 4.1. *If the true model includes a contemporaneous relation driven by a time-invariant network with heterogeneous asset impact, the Fama-McBeth approach based on ordinary least squares (OLS) leads to a consistent estimation of the risk premia even if we estimate the factor loadings using the misspecified traditional factor model. In other words, the OLS estimates of Λ based on $\hat{\beta}$ and $\hat{\beta}^*$ do coincide.*

Proof. This is a consequence of the fact that, if W is time invariant, the least squares estimator applied to the model in Eq. (40) provides a consistent estimate of our model's reduced-form betas, β^* . In other words, $\beta \equiv \beta^*$, and thus the risk premia also coincide. The same result will arise when noticing that the expected returns are functions of the reduced-form betas. An alternative proof might be obtained by generalizing the framework of Kou et al. (2017) by introducing heterogeneous network impact and adapting their proof. ■

A few remarks are now needed. First, the previous result holds in the special case of scalar or heterogeneous network impact with time-invariant networks. Second, the result holds only for the point estimates of the risk premia and only if we estimate the premia by standard least squares. This is due to the fact that, under our model, the covariance of ε_t^* has a known structure that depends on the network-i.e., the matrix W , and on the parameters included in $\mathcal{R}$. This covariance differs from the estimated covariance of ε_t under the multifactor model. Consequently, inference provides, potentially, different outcomes-that is, different standard errors. Furthermore, if we adopt generalized least squares, even the (small-sample) point estimates differ. Finally, the innovations in (44)

are likely affected by cross-sectional dependence due to spatial-i.e., network driven-interactions. In a time-invariant network case, one might use instead the covariance matrix estimator proposed by Conley (1999).

The previous proposition allows us to highlight that, when the network is time invariant, the risk premium estimation is unaffected by the model misspecification-i.e., the absence of network-driver interdependence among assets. Consequently, our model's main contributions are as follows: (i) the ability to disentangle the structural and network-induced factor exposure, (ii) the possibility of capturing correlation in the residuals of traditional linear factor models, and (iii) the possibility of showing that diversification benefits are weaker than expected.

When the network is dynamic, the expected returns are characterized by

$$\mathbb{E}[R_t|W_t] = r_f + \bar{\beta}\Lambda + \sum_{j=1}^{\infty} \mathcal{R}^j W_t^j \bar{\beta}\Lambda = r_f + \beta_t^* \Lambda. \quad (45)$$

We first point out that the reduced-form betas are time varying, being equal to $\beta_t^* = (1 - RW_t)^{-1} \bar{\beta}$, while we have time-invariant risk premia.

From the previous expected return relation, we have that, under our model with time-varying networks, the classical Fama-McBeth approach for risk-premia estimation involves a second-stage cross-sectional regression in which, if we still focus on reduced-form betas, we must account for the fact that those betas are time varying. Despite this, the model structure allows to apply the classical Fama-McBeth approach after a filtering step. In fact, instead of working with reduced-form betas, we might account for the network dependence on the left-hand side of the pricing Eq. (45), and the structural betas on the right-hand side, as follows:

$$(I - RW_t) (\mathbb{E}[R_t|W_t] - r_f) = \bar{\beta}\Lambda. \quad (46)$$

Note that

$$(I - RW_t) (\mathbb{E}[R_t|W_t] - r_f) = \mathbb{E}[(I - RW_t) (R_t - r_f) | W_t], \quad (47)$$

since we are evaluating expected returns, conditionally to the knowledge of the network dynamic. Consequently, based on the first-pass estimates, where we evaluate our returns model augmented with network dependence, we can filter out the network-driven cross-sectional dependence from the left-hand side of the pricing equation, computing the quantities

$$Z_t = (I - RW_t) (R_t - r_f) \quad (48)$$

and then perform the Fama-McBeth approach on

$$Z^e = \hat{\beta}\Lambda + v, \quad (49)$$

where Z^e are time averages of Z_t and $\hat{\beta}$ denotes the estimated structural betas obtained from the model.

We can highlight the impact of our methodology by comparing the estimation of the risk premia, including time-invariant reduced-form betas estimated from a misspecified linear factor model, with the risk premia obtained following our procedure. In principle, the two approaches should lead to different estimates of the risk premia.

However, the evolution of the network also plays a role. In fact, the time-varying network structure lies between two extreme cases, the full connection case where $W_t = \frac{1}{K-1}(\mathbf{1}\mathbf{1}' - I)$ and the absence of connections-i.e., $W_t = 0$. This is a consequence of the spatial interpretation of the matrices W_t and of the normalization rule we adopted.¹⁶ Further, the time evolution of the reduced-form betas, for a given vector of structural betas, is bounded, and oscillates around the given structural betas values. The latter is a consequence of

$$\beta_t = \bar{\beta} + \sum_{j=1}^{\infty} \mathcal{R}^j W_t^j \bar{\beta} \quad (50)$$

combined with the bounded structure of W_t and with the constraints that the parameters in $\mathcal{R}$ must satisfy. Therefore, for an increasing sample size, we postulate that the betas estimated from a classical linear factor model converge to the average of the time-varying reduced-form betas. In turn, the average is also close to the structural beta vector due to the standardization that provides W_t matrices with elements that are, on average, relatively small. Consequently, we expect that the estimation of the risk premium from a misspecified linear factor model and from our methodology, based on network-filtered returns Z_t , will be close. In addition, given that our approach accounts for the time-varying nature of reduced-form betas in the computation of Z_t , we expect differences in terms of the estimators' efficiency. We verify these elements by means of simulations.

5. Simulation analysis

To show the capabilities of the proposed framework and to underline the effect due to model misspecification-that is, neglecting the network links across assets-we include in this section a set of simulations.

¹⁶ We also issue a reminder that the networks and the parameters in $\mathcal{R}$ are such that $I - RW_t$ is always nonsingular.

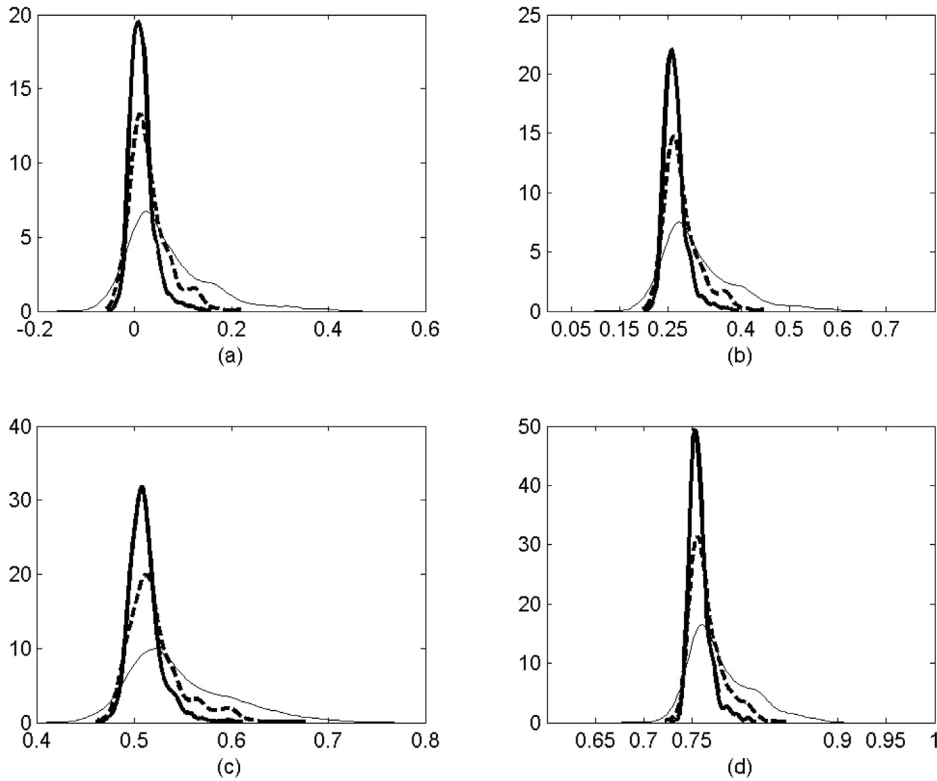


Fig. 1. Distortions of the coefficients ρ under the correctly specified model. True values: (a) $\rho = 0$, (b) $\rho = 0.25$, (c) $\rho = 0.5$, and (d) $\rho = 0.75$. Lines refer to different sample sizes: $T = 200$ is the thin grey line, $T = 500$ is the dashed line, and $T = 1000$ is the thick black line.

5.1. Scalar network impact

First, we concentrate on the simplest network-augmented linear factor model, with time-invariant W and scalar ρ . Such a baseline design will provide some expected results, as we will point out in a few lines. The first data-generating process we consider is a linear factor model with a unique risk factor, a scalar network impact, and a fixed (and known a priori) network matrix W :

$$(I - \rho W)(R_t - \mathbb{E}[R_t]) = \tilde{\beta} F_t + \eta_t, \quad (51)$$

with the following specification for parameters, shocks, and asset interconnections. We consider $K = 100$ assets, thus focusing on a somewhat large cross-sectional dimension, and assume we simulate monthly returns. The coefficient ρ assumes fixed values $\rho \in \{0, 0.25, 0.5, 0.75\}$, allowing us to compare the case of no network impact with different and increasing levels of network impact; note that, when $\rho = 0$, our model collapses to the traditional linear factor model. The factor-loading coefficients are randomly generated from $\beta_i \sim \mathcal{U}(0.8, 1.2)$, $i = 1, 2, \dots, K$, giving positive factor loadings with an average value of 1. We simulate the factor returns from a Gaussian density, $F_t \sim \mathcal{N}(\mu_F, \sigma_F^2)$ with $\mu_F = 0$ and $\sigma_F = 15\%$, on a yearly basis. The risky asset's expected return equals $\mathbb{E}[R_t] = r_f + (I - \rho W)^{-1} \beta \Lambda$, β being the K -dimensional vector of betas simulated above, while the factor risk premium equals 5% on a yearly basis, and the risk-free rate is set to 1% on a yearly basis. The matrix W comes from a simple and naive design: each of its off-diagonal elements is extracted from a Bernoulli density $w_{i,j} \sim B(p_B)$ with $p_B = 0.3$; the simulated W is then row normalized. The shocks are extracted from a Gaussian $\eta_t \sim \mathcal{N}(0, \Omega)$, with Ω being a diagonal matrix with diagonal elements extracted from a uniform $\omega_{i,i}^2 \sim \mathcal{U}(10\%, 25\%)$, with limits referring to a yearly horizon. We simulate 500 sequences of monthly returns with three different sample sizes: $T = 200, 500$, and 1000.

We proceed to the estimation of the model parameters by Concentrated Maximum Likelihood methods. The baseline simulation provides expected results. First, the estimators of the coefficients ρ and of the (structural) vector β have an asymptotically normal density with dispersion decreasing with the sample size (see Table 1). Fig. 1 reports a kernel estimate of the distortion $\hat{\rho} - \rho$ across different values of ρ , while Fig. 2 provides a kernel density for the cross-sectional average (over the assets) of the distortions $\hat{\beta}_i - \beta_i$, $i = 1, 2, \dots, K$; all graphs show the plots for the three different sample sizes. We note that the coefficients converge to the true values and that their dispersion decreases with the sample size, as expected.

If we estimate a standard linear factor model on the series simulated from Eq. (51)-that is, we fit

$$R_t = \gamma_0 + \gamma_1 F_t + \varepsilon_t, \quad (52)$$

Table 1

Mean and standard deviation for the distortions ρ and β under a correct model specification across different values of the network impact and different sample sizes. Values computed across 500 replications.

T	$\rho = 0$		$\rho = 0.25$		$\rho = 0.5$		$\rho = 0.75$	
	Mean	Std.dev	Mean	Std.dev	Mean	Std.dev	Mean	Std.dev
Distortions for ρ								
200	0.067	0.084	0.062	0.075	0.050	0.058	0.028	0.032
500	0.029	0.040	0.028	0.037	0.024	0.030	0.014	0.017
1000	0.015	0.024	0.014	0.023	0.012	0.019	0.007	0.011
Cross-sectional average of the distortions β								
200	−0.072	0.091	−0.088	0.107	−0.096	0.107	−0.119	0.133
500	−0.032	0.043	−0.040	0.053	−0.045	0.061	−0.059	0.072
1000	−0.016	0.026	−0.020	0.032	−0.022	0.034	−0.031	0.045

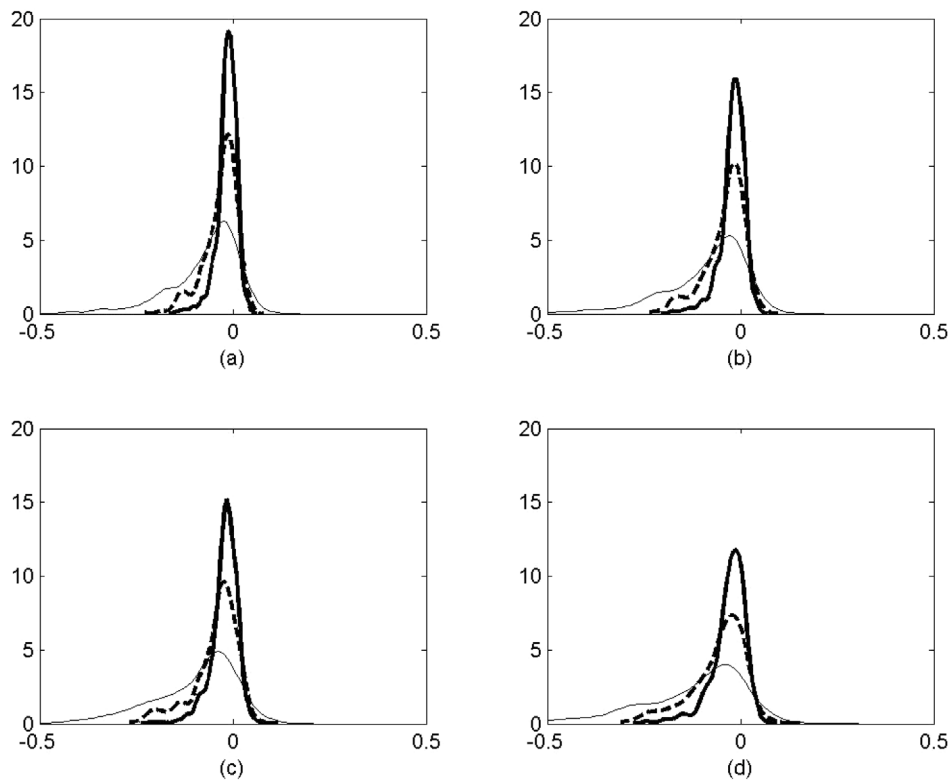


Fig. 2. Cross-sectional average of the distortions for the coefficients β under the correctly specified model across different values of ρ . True values: (a) $\rho = 0$, (b) $\rho = 0.25$, (c) $\rho = 0.5$, and (d) $\rho = 0.75$. Lines refer to different sample sizes: $T = 200$ is the thin grey line, $T = 500$ is the dashed line, and $T = 1000$ is the thick black line.

we have that $\gamma_0 = \mathbb{E}[R_i]$, $\gamma_1 = (I - \rho W)\beta$, and $\mathbb{V}[\varepsilon_i] = (I - \rho W)^{-1} \Omega (I - \rho W')^{-1}$. Therefore, estimating the linear factor model by least squares, we estimate the reduced-form representation of our network-augmented linear factor model. The γ_1 coefficients, by construction, will be larger than the structural coefficients β when we simulate from a data-generating process with positive ρ . This is confirmed by Fig. 3 and Table 2, in which we report the kernel density for the cross-sectional average of $\hat{\gamma}_{1,i} - \beta_i$, $i = 1, 2, \dots, K$ and some descriptive statistics. Moreover, the residuals of the linear factor model will be correlated, with average correlation increasing with ρ (see Table 2).

Fig. 3 and Table 2 confirm that, by fitting a linear factor model, we estimate a beta much larger than the structural value, with distortion increasing with the impact from the network connections. As a consequence, the value of the true and structural factor loading might sensibly differ from the one that is empirically observed, being doubled for values of ρ equal to 0.5, and thus not particularly elevated.

When analysing the residual correlations, we can see that they are zero when the linear factor model is correctly specified—that is, when $\rho = 0$. However, in the presence of a network impact, the residual correlations start drifting away from zero, their values increasing with ρ . On the contrary, under the correct model specification, the residual correlations are almost zero, as expected.

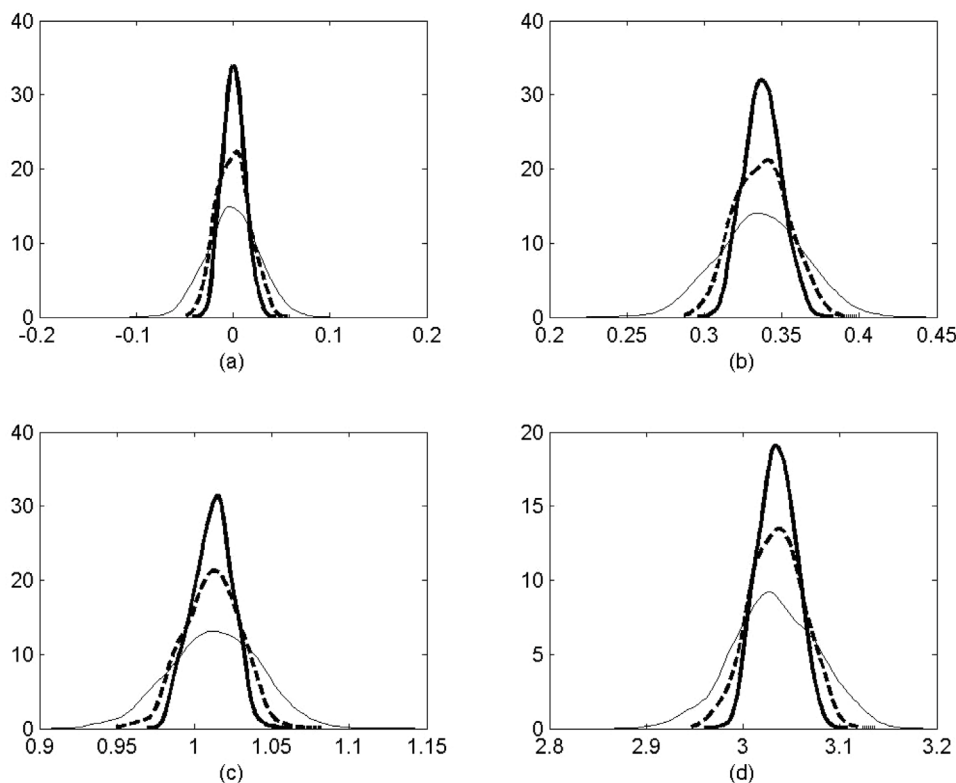


Fig. 3. Cross-sectional average of the distortions $\hat{\gamma}_i - \beta$ under the misspecified linear factor model across different values of ρ for the data-generating process. True values: (a) $\rho = 0$, (b) $\rho = 0.25$, (c) $\rho = 0.5$, and (d) $\rho = 0.75$. Lines refer to different sample sizes: $T = 200$ is the thin grey line, $T = 500$ is the dashed line, and $T = 1000$ is the thick black line.

Table 2

Mean and standard deviation for the cross-sectional average of the distortions $\hat{\gamma}_i - \beta$ under model misspecification, upper panel; average residual correlation under model misspecification, central panel; and under correct model specification, lower panel. Statistics computed across different values of the network impact and different sample sizes. Values computed across 500 replications.

T	$\rho = 0$		$\rho = 0.25$		$\rho = 0.5$		$\rho = 0.75$	
	Mean	Std.dev	Mean	Std.dev	Mean	Std.dev	Mean	Std.dev
Cross-sectional average of the distortions $\hat{\gamma}_i - \beta$								
200	0.000	0.026	0.337	0.027	1.013	0.028	3.033	0.043
500	0.000	0.016	0.337	0.017	1.011	0.018	3.034	0.027
1000	0.000	0.011	0.337	0.012	1.012	0.012	3.035	0.019
Average residual correlations under the linear factor model								
200	0.000	0.003	0.009	0.007	0.034	0.014	0.149	0.035
500	0.000	0.002	0.009	0.007	0.033	0.013	0.149	0.035
1000	0.000	0.001	0.009	0.006	0.033	0.013	0.149	0.035
Average residual correlations under the network-augmented linear factor model								
200	-0.001	0.004	-0.002	0.003	-0.002	0.003	-0.002	0.003
500	-0.001	0.002	-0.001	0.002	-0.001	0.002	-0.001	0.002
1000	0.000	0.001	0.000	0.001	0.000	0.001	-0.001	0.001

Next, we move to the estimation of the factor risk premium. We adopt the widely used two-pass regression approach of [Black, Jensen, Scholes, et al. \(1972\)](#) and [Fama and MacBeth \(1973\)](#). In linear factor models, the first stage corresponds to the estimation of the factor loadings—that is, the betas. Differently, in our model, the first stage is the estimation of reduced-form betas starting from the estimated coefficient ρ and corresponds to a by-product of the concentrated maximum likelihood estimation approach adopted. We stress that, under scalar ρ and with a static W , the reduced-form betas and the linear factor model betas are asymptotically equivalent. The second regression is a cross-sectional one, which takes as dependent the average risky asset returns and regresses them on the estimated betas (reduced-form betas, in our model). As pointed out by [Black et al. \(1972\)](#), the estimated risk premium suffers from an error-in-variable problem and is thus inconsistent. Standard solutions include grouping assets into portfolios,

Table 3

Mean and standard deviation of the estimated risk premia across the 500 replications. The cross-sectional regression adopts an OLS estimator. The true risk premium corresponds to 0.4167 at the monthly frequency.

T	$\rho = 0$		$\rho = 0.25$		$\rho = 0.5$		$\rho = 0.75$	
	Mean	Std.dev	Mean	Std.dev	Mean	Std.dev	Mean	Std.dev
Estimated risk premia from a linear factor model								
200	0.415	0.317	0.417	0.318	0.418	0.318	0.419	0.319
500	0.417	0.198	0.418	0.199	0.418	0.199	0.418	0.199
1000	0.423	0.137	0.423	0.138	0.423	0.138	0.423	0.138
Estimated risk premia from a network-augmented linear factor model								
200	0.418	0.322	0.419	0.323	0.420	0.324	0.421	0.325
500	0.418	0.200	0.419	0.200	0.419	0.200	0.419	0.200
1000	0.423	0.138	0.423	0.138	0.424	0.138	0.424	0.138

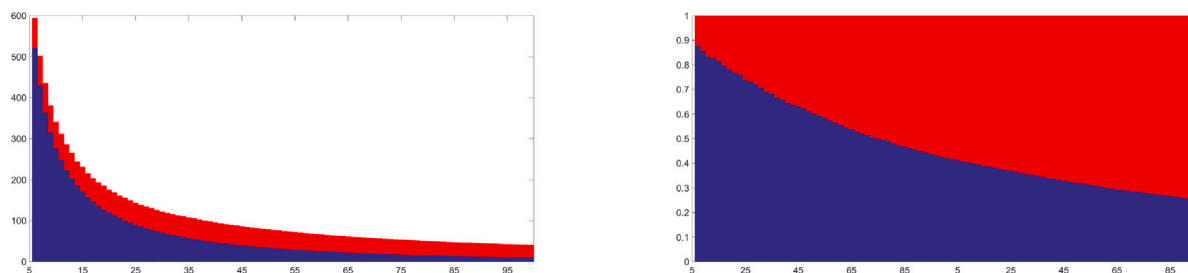


Fig. 4. $1/N$ portfolio idiosyncratic risk components: structural risk (blue) and network-related risk (red) across different portfolio sizes using the same assets adopted in the simulations and with $\rho = 0.5$. Absolute decomposition (upper) and relative decomposition (lower). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

increasing the sample size, and increasing the cross-sectional dimension. We apply the second one, since we are working in a purely simulated setting in which we do not control for the risky asset's *market value*. As a consequence, we expect distortions in the estimation of the risk premia for small sample sizes, and, given the asymptotic equivalence of the betas, no difference between our model and the misspecified linear factor model. However, an expected result like this does not impact on the purpose of our simulation design, as our final objective is not the correct estimation of the risk premia, but rather to highlight the differences in the estimated risk premia obtained from either a network-augmented linear factor model or a (misspecified) linear factor model. We finally point out that the cross-sectional estimation of the risk premium could come from either a standard OLS or a GLS estimator. For the latter, we note that the correct model specification allows for a more precise design of the residuals' covariance (in the reduced-form representation of our model).

In Table 3 we report the estimated risk premia. As expected, the premia are very close to the true value, with a dispersion decreasing in T . The limited distortions depend on the large sample sizes we consider.¹⁷ No difference emerges when we compare the correctly and incorrectly specified models. Finally, we should point out that the OLS and GLS estimators provide substantially equivalent results, and thus we report only the OLS case.

As a further example, we consider the portfolio variance, $1/N$, concentrating on the role played by the idiosyncratic risks. We order assets on the basis of their idiosyncratic risk and decompose the portfolio idiosyncratic risk into the structural component and the network effect. We consider portfolios with N varying from 5 to 100. Fig. 4 reports the decomposition in both absolute and relative terms. Notably, network exposure induces a decrease in the idiosyncratic risks that is much smaller than that associated solely with the structural risks, and with a relative weight increasing over time. Such a result leads to diversification benefits that are reduced compared to the ideal case of independent idiosyncratic shocks (associated with the reduced-form model representation).

To evaluate the impact of the various settings of the data-generating process, we run a number of robustness checks: we simulate the vector β from a Gaussian with mean 1 so that the betas are more concentrated around the mean but also characterized by a larger variance; we increase the volatility of the common factor to a yearly value of 25%; we change the network density by setting $p_B = 0.15$ and $p_B = 0.45$ —or, maintaining the same density, we simulate different networks; we modify the factor risk premium to $\Lambda = 3\%$ or $\Lambda = 10\%$; we increase the relevance of the idiosyncratic shocks by sampling elements of Ω as $\omega_{i,i}^{1/2} \sim \mathcal{U}(20\%, 50\%)$.

None of these changes affects the previously reported results.¹⁸

¹⁷ Similar results have been obtained with smaller samples of 60 and 120 observations.

¹⁸ Additional figures and tables are available upon request.

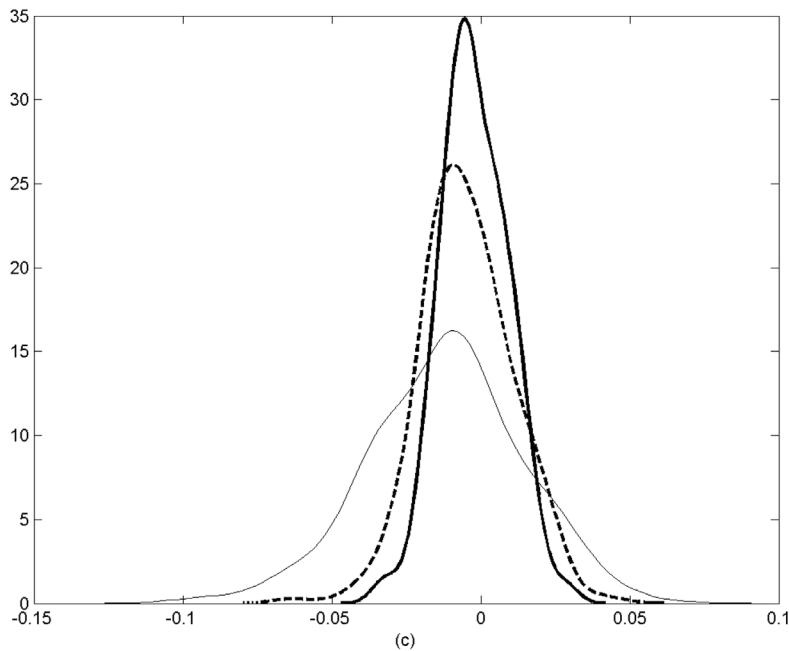


Fig. 5. Distortions of the cross-sectional average of $\text{diag}(\mathcal{R})$ under the network-augmented linear factor model. Lines refer to different sample sizes: $T = 200$ is the thin grey line, $T = 500$ is the dashed line, and $T = 1000$ is the thick black line.

5.2. Heterogeneous network impact

The second simulation design we consider adds the heterogeneity to the network impact. We thus move from the coefficient ρ to the diagonal matrix $\mathcal{R}$. The asset-specific network impact comes from a normal density, $\rho_i \sim \mathcal{N}(0.5, 0.01)$, such that with probability close to 99%, ρ takes values between 0.25 and 0.75. In order to control the computational time, we reduce the cross-sectional dimension for this simulation and set $K = 20$.

For that case, we provide in Fig. 5 a kernel density for the cross-sectional average of $\hat{\rho}_i - \rho_i$, $i = 1, 2, \dots, K$ for different sample sizes.

We do not present further results for the estimated factor loadings and residual correlations associated with the fit of the standard linear factor model because they provide the same evidence as in the first simulation design: the betas are larger than the structural values and the residuals are correlated. We only point out that, in the presence of heterogeneity in the network impact, the residual correlations are even higher than in the case of scalar ρ .¹⁹

Instead, we provide in Table 4 further evidence from the risk premium estimation. Notably, the estimated risk premia present a slight distortion (overestimation) in relation to the previous simulation design. We link them to the introduction of the asset-heterogeneous impact of the network, which intuitively amplifies the impact of the error-in-variable problem. Increasing the sample size, the distortions tend to decrease, as does the dispersion of the estimated risk premia. There are no differences found between the two estimation approaches, as in the previous case. Finally, as expected, the correctly specified and misspecified models provide comparable results. We stress that this is a consequence of the data-generating process we follow, in which the risk premium is estimated by looking at the reduced-form betas. In the current data-generating process, with heterogeneous network impacts, the linear factor model provides consistent estimates of the reduced-form betas, but does not allow separation of the network and structural elements that affect the betas.

For the first design, we provided an example associated with the decomposition of the equally weighted portfolio's idiosyncratic risk, with weights equal to $1/N$, into the standard component and the network-related component. Here, we repeat the same exercise with two different $\mathcal{R}$ matrices: the first is the one used above, while the second also allows for the presence of negative ρ_i coefficients in half of the simulated assets. This second example allows us to highlight the *risk-absorption* effect of the network exposure. While for the first case, the results are qualitatively similar to those of the scalar ρ case, when we introduce negative ρ_i values, and order assets in descending order according to their ρ_i values, we note that the introduction of assets with negative ρ_i s leads to a moderate decrease of the interconnection impact on the idiosyncratic risk (the fourth component of the variance decomposition). Such an effect could even become negative, thus leading to the absorption of risk by the linked assets-or, in other words, to the amplification of the diversification benefits. This is evident in Fig. 6, where we report the contribution of the fourth component,

¹⁹ Additional tables and figures are available upon request.

Table 4

Mean and standard deviation of the estimated risk premia across the 500 replications. The cross-sectional regression adopts an OLS or GLS estimator. The true risk premium corresponds to 0.4167 at the monthly frequency.

T	Linear factor model				Network-augmented linear factor model			
	OLS		GLS		OLS		GLS	
	Mean	Std.dev	Mean	Std.dev	Mean	Std.dev	Mean	Std.dev
200	0.440	0.326	0.440	0.324	0.443	0.331	0.442	0.328
500	0.443	0.201	0.443	0.200	0.444	0.202	0.444	0.201
1000	0.429	0.149	0.429	0.148	0.429	0.149	0.430	0.147

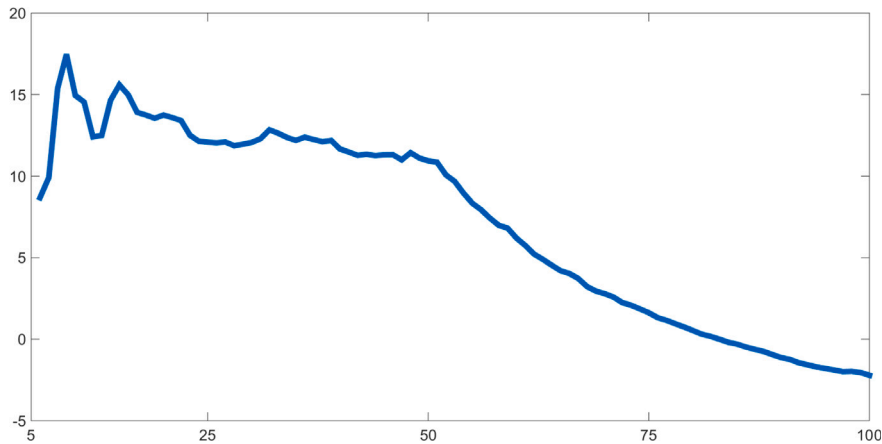


Fig. 6. Fourth component of the equally weighted portfolio variance with weights equal to $1/N$ - impact of interconnections on idiosyncratic risk. Assets in descending order of ρ_i values.

the network effect on the idiosyncratic component of the equally weighted portfolio variance, with weights equal to $1/N$, where the portfolio size increases from 5 to 100 assets, and assets are ranked in descending order of ρ_i . The last 50 assets have negative network impacts, and the contribution of the interconnections to the idiosyncratic risk becomes negative around asset 80.

5.3. Dynamic W and heterogeneous network impact

In the third simulation design, we combine the heterogeneity in the asset-network impact with the time change in the network connections across assets. Now the data-generating process is

$$(I - RW_t)(R_t - \mathbb{E}[R_t]) = \tilde{\beta}F_t + \eta_t, \quad (53)$$

Note that here, unlike in the previous designs, the expected returns, conditional on W_t , are dynamic. To generate a time change in the W_t , we chose a simple approach, starting from the empirical evidence that the links between assets are persistent; that is, we do not have networks that are completely different at times t and $t + 1$. Moreover, as noted in the previous section, we do not allow for a change in W_t at every t , but rather modify W_t every $m = 20$ observations.

We change W_t according to the following scheme: at time 1, we sample W_1 as in the first design; that is, the off-diagonal elements $w_{i,j} \sim B(p_B)$ with $p_B = 0.3$; every m observations, each off-diagonal $w_{i,j}$ can take one of only two values, 0 or 1, and is driven by a Markov chain, with diagonal elements of the transition matrix set as $p_{00} = p_{11} = 0.9$. Such a choice ensures persistence in the W_t , with possibly long-lasting increases/decreases in the associated network density. Finally, we point out that the W_t matrices have been normalized with the maximum row normalization.

We now present a number of results obtained from this simulation design. First, we focus on the coefficients ρ_i . As in the previous case, Fig. 7 reports the kernel density for the average of $\text{diag}(\hat{R}) - \text{diag}(R)$ for different sample sizes. We observe a convergence (on average) of the concentrated estimates to the true values with increasing sample sizes, as expected.²⁰

Secondly, we note that, with the data-generating process shown in Eq. (53), the linear factor model does not estimate the reduced-form betas as, by construction, they are time varying: $\gamma_1 \neq (I - RW_t)^{-1}\beta$. Therefore, to evaluate the distance between those two values, we compute the distortions $\hat{\gamma}_1 - (I - RW_t)^{-1}\beta$ and compare them to the distortions under the correctly specified model $(I - \hat{R}W_t)^{-1}\hat{\beta} - (I - RW_t)^{-1}\beta$; in both cases, we focus on the cross-sectional averages of the distortions. We collect the results in

²⁰ Detailed tables with coefficient-specific results are available upon request.

Table 5

Mean and standard deviation of the cross-sectional averages for the distortions between the estimated betas under the misspecified linear factor model and the reduced-form betas induced by the true model (left-side columns), and between estimated and true reduced-form betas under the correctly specified model (right-side columns).

T	Linear factor model		Network-augmented linear factor model	
	Mean	Std.dev	Mean	Std.dev
200	0.153	0.008	0.082	0.014
500	0.205	0.003	0.055	0.008
1000	0.194	0.002	0.040	0.006

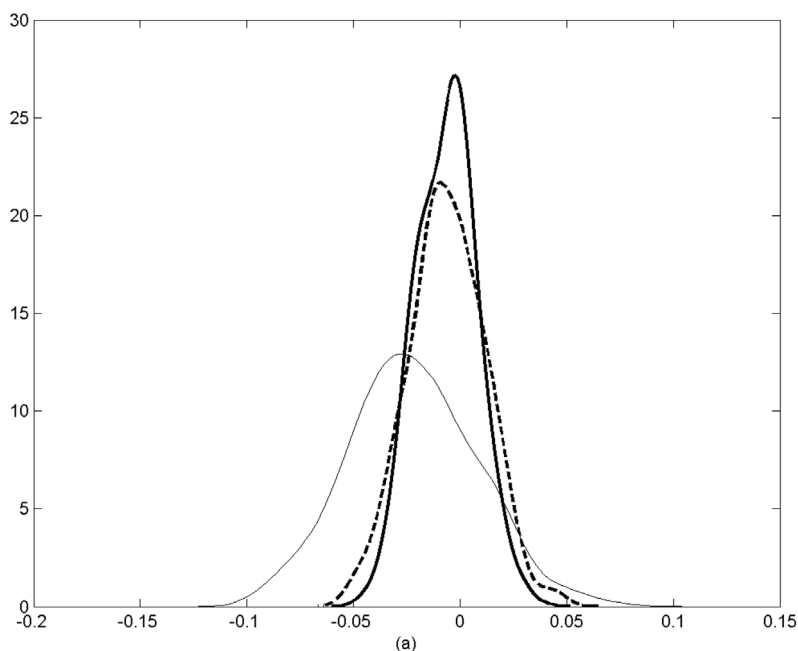


Fig. 7. Distortions of the cross-sectional average of $diag(\mathcal{R})$ under the network-augmented linear factor model. Lines refer to different sample sizes: $T = 200$ is the thin grey line, $T = 500$ is the dashed line, and $T = 1000$ is the thick black line.

Table 5. The table shows that the correctly specified model captures the evolution of the reduced-form betas, which, we recall, are conditional on the knowledge of the network links. Moreover, the distortions decrease in both mean and dispersion. In contrast, for the misspecified model, the distortions do not clearly converge towards the true values, but seem to be characterized by an average overestimation of the factor impact.

Finally, we move on to the risk premium estimation and report the results in [Table 6](#). In this case, the GLS estimator we adopt for the correctly specified model takes into account the known time-varying nature of the reduced-form betas. The risk premium estimation is carried out by focusing on the excess returns filtered from the network dependence, and on the structural betas. We first highlight that the OLS and GLS estimates are substantially equivalent. Thus, there is no effect associated with the estimator adopted. Then we come to the most interesting finding: the risk premia are very close to the true values for both the correctly and incorrectly specified models, and similarly the risk premium dispersions are very close together under the two estimated models. Distortions were somewhat expected, but they have been cancelled out by two elements. The first is the introduction of an averaging across the different matrices W_t . In fact, under the linear factor model, we estimate the betas using the entire sample, which is implicitly affected by the various networks. The reduced-form model estimators are implicitly averaging across the W_t . The second element is the pattern characterizing the matrix W_t , which is not exploding. Nevertheless, this second element plays a minor role. Further simulations with different dynamics for the W_t , introducing a linear or exponential increase in the network density, or a level shift in the network density, confirm the finding.²¹

As we anticipated in the previous section, the most striking result is the reduction in the risk premium's standard errors. Given the relevant temporal dimension we adopt, the risk premium estimation on a misspecified linear factor model is equivalent to the

²¹ Additional results are available upon request.

Table 6

Mean and standard deviation of the estimated risk premia across the 500 replications. The cross-sectional regression adopts an OLS or GLS estimator. The true risk premium corresponds to 0.4167 at the monthly frequency.

T	Linear factor model				Network-augmented linear factor model			
	OLS		GLS		OLS		GLS	
	Mean	Std.dev	Mean	Std.dev	Mean	Std.dev	Mean	Std.dev
200	0.418	0.065	0.418	0.062	0.414	0.018	0.415	0.016
500	0.419	0.044	0.419	0.045	0.416	0.010	0.416	0.010
1000	0.418	0.030	0.418	0.030	0.416	0.005	0.416	0.005

one obtained from our approach, but the two models lead to completely different dispersions of the estimates. Taking the network dependence into account, the standard errors reduce, on average, by more than 75%. Such evidence might have a significant impact on the evaluation of the statistical significance of the risk premia recovered from real data.

6. Empirical analysis

To highlight the features of our methodology we conduct an empirical analysis by using the Input–Output (I/O) requirements tables compiled and maintained by the U.S. Bureau of Economic Analysis. The I/O tables are designed to show the level of industry gross output that is required to produce a specified level of final uses.²² In particular, the Input–Output table we use is the so-called industry-by-industry total requirements table showing the production required, both directly and indirectly, from a given industry per dollar of delivery to a different industry.²³ We download the yearly industry-by-industry total requirements tables from 2004 to 2017. We focus on tables aggregating companies into 15 macroeconomic sectors associated with the NAICS two-digits codes.²⁴

We first recover a collection of spatial matrices from the I/O requirements table. In our framework, the I/O matrix can be useful in understanding to which extent the output of an industry contributes to the input of other industries. Therefore, these tables permit to understand in which way a productivity shock generated in a specific industry can spread within the economic system. In our choices, we follow Acemoglu et al. (2012) arguing that, in the presence of intersectoral input–output linkages, microeconomic idiosyncratic shocks may lead to aggregate fluctuations.²⁵ Productivity shocks propagate across the intersectoral links, and changes of I/O affect the overall economy since the strength among links changes. Herskovic (2018), using a I/O matrix in his general equilibrium model, shows that network concentration (output share), sparsity (strength links among nodes) and productivity (that arises endogenously) are relevant characteristics affecting the stochastic discount factor. Sparsity is directly related to productivity gains due to sectors' connectivity; in other words, the network defines the shape of the production function. When network sparsity increases, firms optimize inputs allocation based on changes in their marginal productivity, substituting inputs that have lower marginal product with those that have higher marginal product. The network weights represent the output elasticity with respect to different inputs, and the dispersion of these output elasticities increases when sparsity increases. Innovations in network sparsity carry a positive price of risk: stocks positively correlated with sparsity have higher returns because investors need compensation for holding this risk.²⁶ A productivity shock would affect the topology of the I/O matrix and, consequently, have an impact on the stocks returns. The idea is simple, a shock to a single firm (or sector) could have a much larger impact on the macroeconomy if it reduces the output of not only this firm (or sector), but also of others that are connected to it through a network of input output linkages; see also Acemoglu, Akcigit, and Kerr (2016). Further, Ozdagli and Weber (2017) show that the input–output industry-by-industry structure of the U.S. economy is an important of an important propagation channel of aggregate monetary policy shocks.

As an illustrative example, consider the case of an exogenous shock that hits the manufacturing sector, causing, for instance, a production drop. This shock could trigger a contraction in the agriculture or construction sector because the industry output is not sufficient to satisfy their demand of industry goods. The network structure will affect the intensity of the contraction, with thicker edges we will have a more intense contraction. In turn, these shocks might lead to negative effects to the equity prices of both the industrial companies as well as of companies belonging to the agriculture and construction sectors. If the shock origin is not related to global macroeconomic conditions, the production contraction might not be directly observable and might not be properly proxied by common factors, such as the market index.

Based on these considerations, we assume that the network is exogenously determined and that its characteristics define the mechanism associated with endogenous aggregate productivity shocks. Differently from Kou et al. (2017), we do not impose a network based on physical distances but use an I/O matrix. In addition, since firms adjust their input preferences according to input marginal costs, marginal product impact on their production function and input spot prices I/O matrix changes over time.

²² A detailed description of the tables and or their content is available at https://apps.bea.gov/papers/pdf/IOmanual_092906.pdf

²³ The Input–Output tables are two-way tables are report on the row the industries producing goods and on the column the industries receiving goods.

²⁴ Data are available at <https://apps.bea.gov/itable/index.cfm> The NAICS classification is defined in the Bureau of Economic Analysis website <https://www.bea.gov/sites/default/files/2018-04/2017-industry-code-guide.pdf> and United States Census Bureau <https://www.census.gov/eos/www/naics/>. The availability of SIC codes within the CRSP database prevent for using data prior to 2004.

²⁵ The authors use commodity-by-commodity direct requirements table.

²⁶ If the sparsity increases, the input–output linkages are rearranged, increasing aggregate consumption and decreasing marginal utility, further details are in Herskovic (2018)

Table 7

The table reports the summary statistics for Fama and French (1993) and Carhart (1997) risk factors returns, over 168 months from January 2005 to December 2018. The mean and the standard deviation are annualized. The last column reports the test for evaluating the statistical significance of the mean.

	Observations	Mean	Standard deviation	Minimum	Maximum	Median	Kurtosis	Skewness	t-statistic
Mkt-RF	168	0.074	0.142	−0.172	0.114	0.010	4.936	−0.749	1.935
SMB	168	0.006	0.080	−0.047	0.061	0.000	2.691	0.247	0.287
HML	168	−0.009	0.088	−0.111	0.083	−0.002	5.553	0.031	−0.364
Mom	168	0.016	0.153	−0.344	0.125	0.003	24.304	−2.936	0.383

If the network characteristics are a latent factor, a standard multifactor model would not capture these relations among industries, thus leading to distortions that would be displayed as non-null correlations on the idiosyncratic covariance matrix. On the contrary, the introduction of input–output relationships across economic sectors and the presence of industry specific (ρ_i) spatial parameters, introduces additional flexibility to the standard multifactor model, allowing for the detection of local or spatial dependences among equity returns.²⁷ The higher the ρ_i parameter the higher would be the expected return associated with asset i , everything else being equal. Observing statistically significant and positive ρ_i would mean that a part of the factor exposure of the industry i to global factors would be indirectly induced by the connections with neighbour industries. Note that the I/O tables also include intra-industry input–output.

To convert input–output tables into matrices representing interconnections among sectors, we remove the tables diagonal (the intra-industry input–output) and use the normalization rule in Eq. (25). This gives us a collection of 14 annual networks (from 2004 to 2017) that will lead to a time-varying component in the simultaneous relations among asset returns; see Eq. (24).

Beside the matrices representing the interdependence between economic sectors, we download a set of common factors, namely, the market, size, book-to-market, and momentum factors available at the Kenneth French website; see Fama and French (1993) and Carhart (1997) for details on the risk factors' construction. We recover factors at the monthly frequency, and we will estimate the model parameters, that is, the structural parameters associated with the risk factors and the sector-specific spatial parameters ρ_i included in the matrix $\mathcal{R}$, using monthly data. Table 7 reports the summary statistics for the risk factors' returns in the sample spanning January 2005 to December 2018, for a total of 168 observations. In this time horizon, risk factors have a modest performance (not statistically significant); this is due to the impact of both the US subprime and Eurozone sovereign crises. The Momentum is the risk factor with the highest standard deviation, all risk factors are leptokurtic, and Market excess returns and Momentum have negative skewness.

Finally, we build a collection of equity indexes matching the 15 economic sectors represented in the I/O tables. To that purpose, we first download from CRPS the monthly stocks returns for all listed firms on AMEX, NYSE, NASDAQ. Then, we filter the data according to Fama and French (1993) and we group the stocks into Equally-Weighted (EW) and Valued-Weighted (VW) portfolios according to the NAICS two-digit classification. We note that the CRSP database contains NAICS codes at the single security level only from August 24, 2001 onwards. Consequently, the NAICS code availability limits the time span of our analysis to the range 2005–2018.

Notably, while the sector indices and the factors have a monthly frequency, we will make use of yearly I/O-based networks covering the years from 2004 to 2017. Fig. 8 shows the sectors interconnections in 2017.

This is coherent with the fact that asset interconnections do not vary with a high frequency, in particular when they represent I/O-based relations that would require, at least some month, to show evidence of changes. In addition, we believe it is reasonable to introduce a lag between the spatial matrix W and the monthly asset returns because a shock in the real economic does not have contemporaneous effects on the equity returns. Consequently, for the monthly returns in a given calendar year, the network-augmented linear factor model capture asset interconnections by using the I/O network of the previous calendar year. As a side issue, the introduction of a yearly lag for the network also avoids possible endogeneity issues; this helps in satisfying, weakly, one of the fundamental assumptions in the spatial econometrics literature, that is, the exogeneity of spatial matrix W .

6.1. Results

We proceed to the estimation of the network-augmented linear factor model, or spatial model, following the concentrated maximum likelihood approach as in Eq. (34). Then, we compare our model with the four-factor model of Carhart (1997), which represents our benchmark.²⁸

Table 8 reports the estimated spatial parameters ρ_j , $j = 1, 2, \dots, 15$. We allow for asset-specific parameters, both in terms of spatial dependence and exposure to common factors. A positive spatial dependence, mediated by the estimated time-varying networks, indicates that the network effect has a positive spillover on that asset. The sectors having the highest and most significant spatial effect are *Retail Trade* (RT) and *Information*(Inf) with coefficients equal to $\rho = 0.53$ and $\rho = 0.47$, respectively. According to

²⁷ ρ_i could be also interpreted as a covariation to the network sparsity.

²⁸ Detailed results for all estimated models are available from the authors upon request.

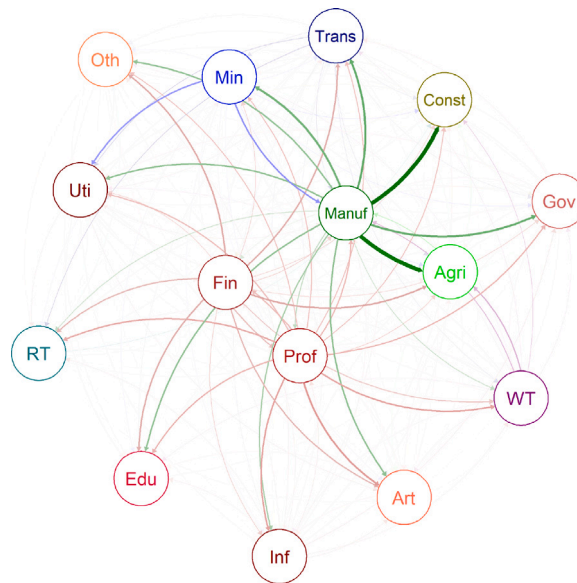


Fig. 8. I/O matrix visualization in 2017. The figure represents the I/O matrix row-normalized according the Eq. (25). Force-directed graph algorithm (Fruchterman & Reingold, 1991) is used to draw the network. The intensity of the arrow (transparency and width) indicates how much sector i needs from j to produce its final output.

Table 8

The table reports spatial parameters ρ from January 2005 to December 2018 relative to 15 sectors' industry portfolios.

Symbol	Sector	ρ
Agri	Agriculture, forestry, fishing, and hunting	0.37***
Min	Mining	0.57
Uti	Utilities	0.06
Const	Construction	0.11
Manuf	Manufacturing	-0.12
WT	Wholesale trade	0.41***
RT	Retail trade	0.53***
Trans	Transportation and warehousing	-0.02
Inf	Information	0.47***
Fin	Finance, insurance, real estate, rental, and leasing	0.04
Prof	Professional and business services	0.21
Edu	Educational services, health care, and social assistance	0.26
Art	Arts, entertainment, recreation, accommodation, and food services	0.22***
Oth	Other services, except government	-0.27
Gov	Government	0.33***

*** Pvalue $\leq 1\%$, ** $1\% < \text{Pvalue} \leq 2.5\%$, * $2.5\% < \text{Pvalue} \leq 5\%$.

Fig. 8, RT and Inf are sectors that lied on the periphery having net balance in favour of incoming links.²⁹ The parameter ρ , known as spatial correlation, measures the impact coming from the other sectors for which the sector is connected to. By looking at Table 8 a first consideration is that ρ s are significant for those sectors for which their margins are not related to the production of goods as RT,³⁰ Inf, Wholesale Trade (WT), Education (Edu) and Agriculture, forestry, fishing and hunting (Agr). The model captures the residuals correlations that are not filtered by common factors and its aggregation through the network links. This evidence is in line with the theoretical prediction regarding the discount factor of Herskovic (2018). Consequently, the relations with their neighbours induce an additional risk, an indirect component that inflates the structural exposure to the common factors.

One of the advantages of our model is its ability to combine the structural exposure to common risk factors with the impact coming from the network connections. Building on this aspect, the model is able to separate the traditional linear factor model

²⁹ This visualization does not include self loops (the diagonal elements of the I/O matrix) because for definition $w_{i,i} = 0$. Therefore the representation considers only the indirect relations among sectors.

³⁰ The margin of Retail Trade is measured as retail sales less the cost of goods sold plus the taxes collected by retail trade establishments.

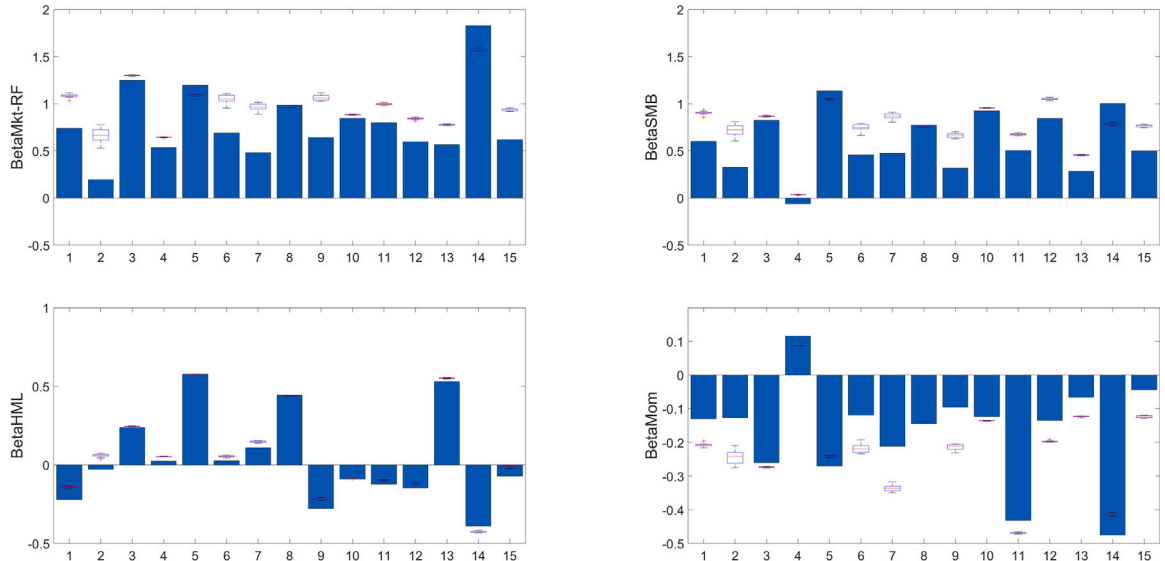


Fig. 9. The figure reports the betas decompositions according to Eq. (11). The blue bar defines the time invariant structural beta $\bar{\beta}$. A box plot is necessary to plot the reduced form beta time varying β_t^* for each sector. The top panel exhibits the factor loading for the Market excess returns $MKT - R_f$. Proceeding in top-down sense the figure shows respectively the factor loading on Size factor SMB , the Book-to-market factor HML and the Momentum factor MOM . At each numbers on the x-axes corresponds the sector, particularly: 1 'Agriculture, forestry, fishing', 2 'Mining', 3 'Utilities', 4 'Construction', 5 'Manufacturing', 6 'Wholesale trade', 7 'Retail trade', 8 'Transportation and warehousing', 9 'Information', 10 'Finance, insurance, real estate', 11 'Professional and business services', 12 'Educational, health and social care', 13 'Arts, entertainment and food', 14 'Other services, except government', 15 'Government'.

betas into two components: first, a structural exposure to systematic risks and second, the exposure to risk coming from the spatial interactions, i.e. the neighbours. This has been shown in Eq. (11) with a constant spatial matrix W . However, with a smoothly evolving matrix we have the following decomposition

$$\tilde{\beta}_t^* = \bar{\beta} + \sum_{j=1}^{\infty} \mathcal{R}^j W_t^j \bar{\beta} \quad (54)$$

$$= \bar{\beta} + \tilde{\beta}_t^{Net} \quad (55)$$

where $\tilde{\beta}_t^*$ is the time-varying reduced form beta matrix. The latter is given as the sum of a time-invariant, structural term, $\bar{\beta}$, and a time-varying component, $\tilde{\beta}_t^{Net}$, influenced by the time-evolution of the spatial matrix W_t . While the structural component is directly obtained from the model estimation, the reduced form betas can be computed only conditionally to the estimated model coefficients. Further, given that the networks change on a yearly basis, we obtain time-varying reduced form betas that change with a yearly frequency. Therefore, while the sign of the network impact is only driven by the $\mathcal{R}$ matrix, i.e. the ρ_j coefficients, the size of the network impact depends both on those parameters as well as on and the network structure implicit in W_t and, obviously, on the structural betas of the neighbour assets. Consequently, with positive ρ_j coefficients, the inflation in the factor exposures might be amplified by the network structure. Fig. 9 exhibits the sectors exposures on common factors decomposed into structural and reduced form exposure. The blue bar represents the structural beta, invariant over time: the pure exposure on the common factors without the network effect. The reduced form factor exposures, β_t^* , are represented by the box plots. The top left panel on Fig. 9 represents the factor loading on the Market. The structural betas are all positive and the difference with the reduced form beta β_{jt}^* defines the network impact which can amplify or reduce the whole factor exposure. If we focus on the sector *Information*, we note it has a structural beta $\bar{\beta}_{MKT}$ equal to 0.64 but the connections with its neighbours inflate its the exposure on the common market factor to an average value of 1.06. The sector with the maximum structural factor loading is *Other services, except government*, with a value equal to 1.82. Nevertheless, in this case the network acts as risk absorber because the reduced form β_{MKT}^* reduces to average value of 1.57. In the same fashion, we analyse the exposures on the remaining factors. On the top right panel of Fig. 9, we observe that the $\bar{\beta}_{SMB}$ are, on average, positive for all sectors except for *Construction*, characterized by a value equal to -0.06 . The structural factor loading on HML, ($\bar{\beta}_{HML}$), changes broadly across the sectors, and reach the greatest value in the case of *Manufacturing*, (0.57), Fig. 9 bottom left. On the other side, *Information* with a $\bar{\beta}_{HML}$ equal to -0.27 reaches the minimum value across the sectors. Interestingly, in this case the network impact shrinks the factor exposure, the β_{HML}^* has a mean value equal to -0.21 because, even though the associated ρ is positive, the structural beta is negative, and the effects are opposite. Finally, on the bottom right panel, the $\bar{\beta}_{MOM}$ have negative values except for *Construction*, with a value of 0.11. *Other services, except government* with a coefficient equal to -0.47 has the minimum structural beta.

To verify whether the spatial model leads to more whitened residuals compared to the four-factor model, we compute the correlations among the idiosyncratic errors for both the Carhart (1997) model and the network-augmented linear factor model;

Table 9

The table reports, respectively, the average, the standard deviation, and the skewness, the absolute average among the residuals correlation for the multifactor and the spatial model. The last row displays the number of elements in the cross-correlation matrix having absolute values lower than 0.1.

	Multifactor model	Spatial model
Average correlation	0.102	0.032
Standard deviation of Correlations	0.151	0.145
Skewness	0.975	0.836
Absolute average correlation	0.132	0.105
Absolute elements <0.1	51.43%	61.90%

Table 10

The table reports the premium computations by using the two-steps (Fama & MacBeth, 1973) procedure corrected by Shanken (1992) respectively for the multifactor and the spatial model.

	Multifactor model		Spatial model	
	λ	p-value	λ	p-value
Mkt-RF	1.024	0.310	1.022	0.004
SMB	−0.438	0.690	−0.453	0.136
HML	−0.662	0.444	−0.637	0.015
MOM	−0.118	0.910	−0.161	0.687

Table 9 summarizes the results. The spatial model has an average residuals correlation of 0.03, lower than the average correlation we observe in the four-factor model case, equal to 0.1. Moreover, the residual correlations distribution in the spatial model has a lower standard deviation 0.14 and skewness 0.84 than the multifactor model case, whose corresponding figures are 0.15 and 0.97, respectively. Therefore, the introduction of a spatial component leads to a contraction in correlation dispersion, decreases the average correlation among residuals and, overall, provides a better fit.

Finally, we compare the risk premia we estimate by the Fama and MacBeth (1973) procedure (and the inclusion of the correction proposed by Shanken (1992)). We estimate the risk premia for both the four-factor and the spatial model. In the latter case, we follow the modified approach described in Section 4, Eq. (47): we first filter out the network-based cross-sectional dependence, and then, on the adjusted returns, we apply the two-step approach of Fama and MacBeth (1973). Table 10 reports the market price of risk for both models. We note that the coefficients have the same signs and almost similar magnitudes. However, the statistical significance is different and much higher in the spatial model case. This result is coherent with the simulation evidence in Table 6, where we observe a standard error reduction in the risk premia estimated by a network-augmented linear factor model.

7. Conclusions

In this paper, we propose a variation of the linear factor model in which networks are used to infer the exogenous and contemporaneous links across assets; we call the model the network-augmented linear factor model.

Building on the model structure, we first show how the presence of network links allows for decomposition of the single-asset (or portfolio) returns, highlighting the impact of interconnected assets in explaining the exposure to common factors. Further, we show how to decompose the risk for assets or portfolios into four components: (i) the systematic component, (ii) the idiosyncratic component, (iii) the impact of the asset interconnections on the systematic risk component (that is, the contribution of network exposure to the systematic risk component), and (iv) the effect of interconnections on the effect of idiosyncratic risk on the systematic risk component (that is, the amplification of idiosyncratic risks that generates systematic/nondiversifiable risk). Starting from this decomposition, we demonstrate the impact of network links on diversification benefits.

Our network-augmented linear factor model allows for the heterogeneous impact of network links on the assets, and also allows for the presence of time variation in the networks. Using the most general model representation, we discuss estimation approaches for recovering model parameters and, more interestingly, we show how to estimate the factor risk premia.

We complement our analyses with a simulation study that shows how the misspecification of the network structure, and the consequent estimation of a classical linear factor model, leads to correlated model residuals and does not allow for recovery of the true, or structural, exposure to common factors. Moreover, in the misspecified linear factor model, the estimation of the risk premia, despite being unbiased (in simulations), is much less efficient compared to the estimates obtained from our methodology.

The empirical application shows that a network augmented linear factor model contributes to interpretation of the factor loading of asset returns, in particular, the contemporaneous relations can interfere with the traditional risk interpretation supported by the standard models. Besides, as shown in the simulations, if a time-varying network is present, then the risk premium estimated following our procedure are more efficient. Results on real data support our claim.

The network-augmented linear factor model is relevant for investors as well as policy-makers and regulators since it helps in understanding whether and to what degree network connectivity has an impact on risk premia, volatilities, diversification, and spillovers between financial assets.

CRedit authorship contribution statement

Monica Billio: Supervision, Methodology. **Massimiliano Caporin:** Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, Supervision. **Roberto Panzica:** Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing. **Loriana Pelizzon:** Writing – original draft, Writing – review & editing, Supervision.

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